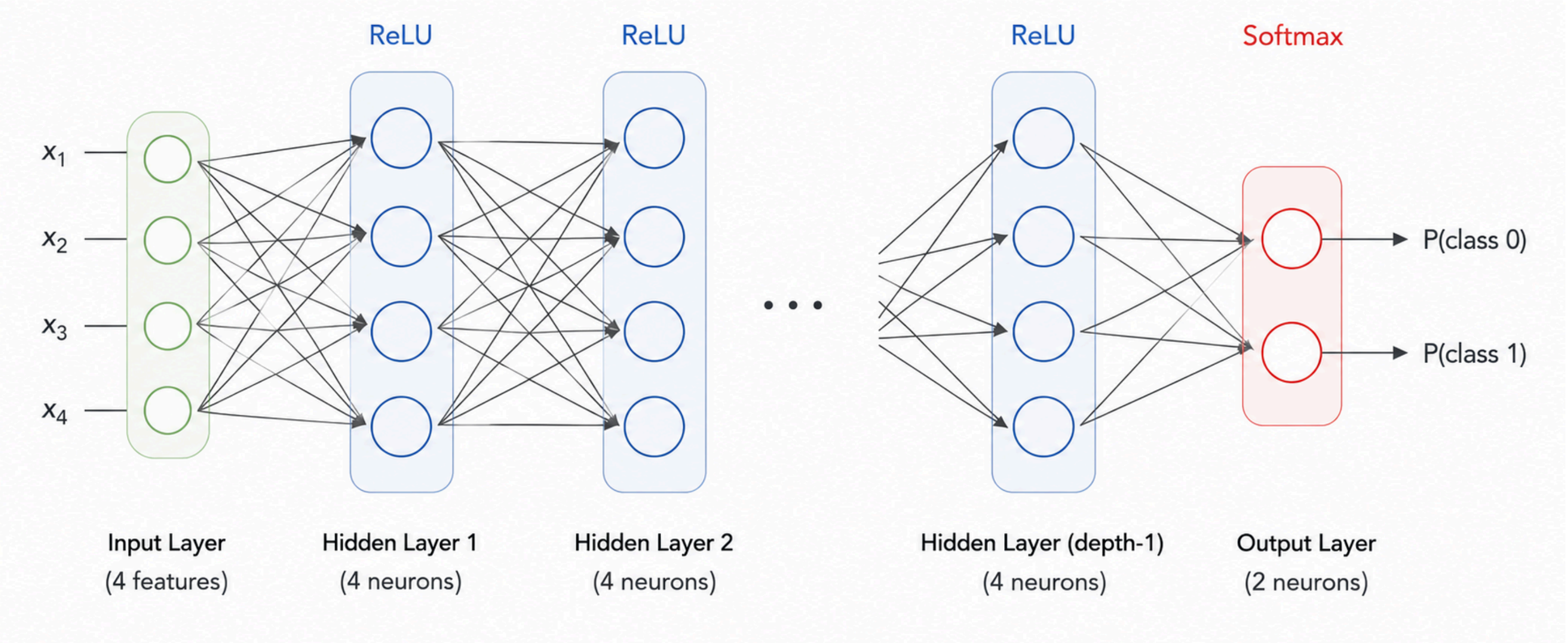
The Power of Quantum Neural Networks

Giacomo Bregolin
Leonardo Fardin
Lara Fogarolo
Francesco Peron

Classical model

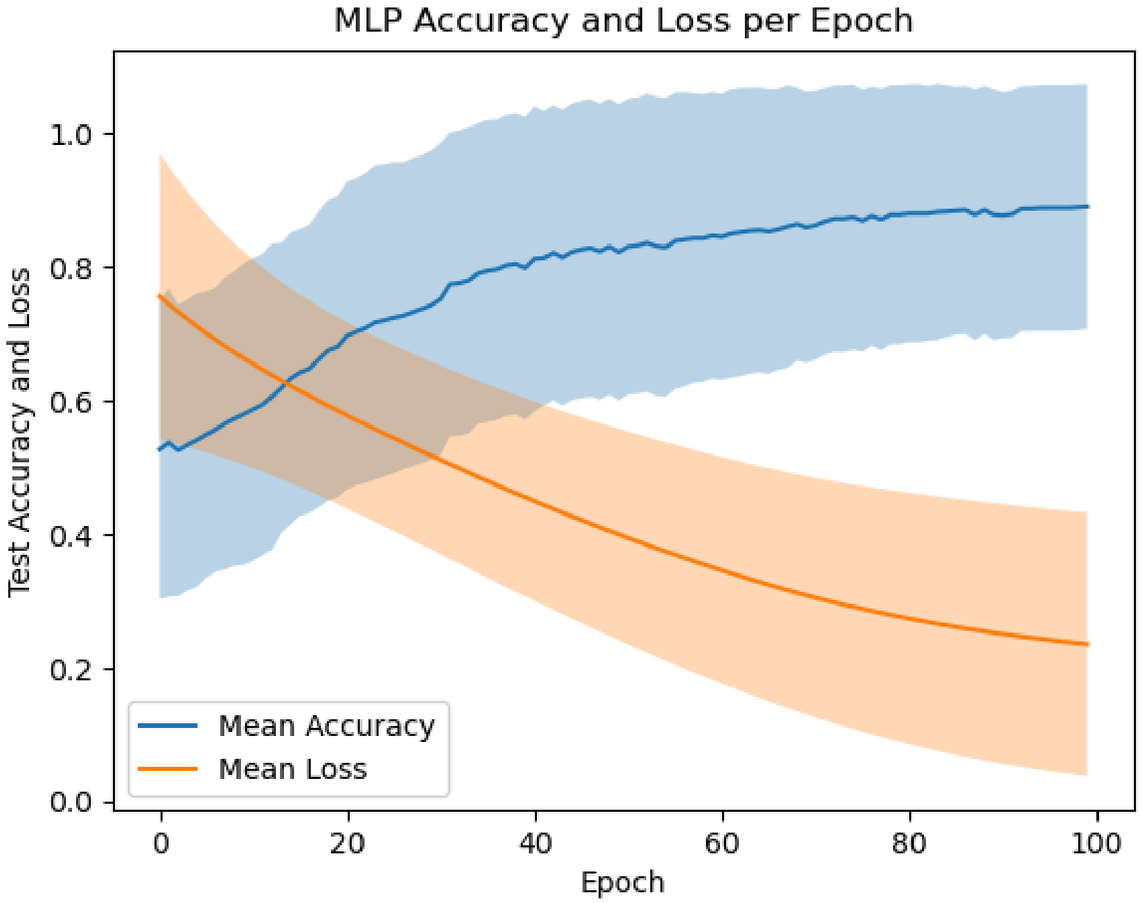
Multi layer perceptron for binary classification



Cross-entropy loss function L: -log(p) where p is the probability assigned to the correct class

In total: 60 learning parameters

Performance of the model



Fisher information

Definition: $$F(\theta) = \mathbb{E}_x[\nabla_\theta L(x, \theta), \nabla_\theta L(x, \theta)^T]$$

We compute it as: $$F = \frac{1}{N} \sum_i g_i g_i^T$$ where $$g_i = \nabla_\theta L(x_i, y_i, \theta)$$

for every trainable parameter

It tells the importance of every paramater of the network

Why Fisher information

- It measures the sensibility of L in changing model parameters
- Computing its eigenvectors we have the directions of the curvature of the loss function
- Eigenvectors with greater eigenvalues are the most informative direction

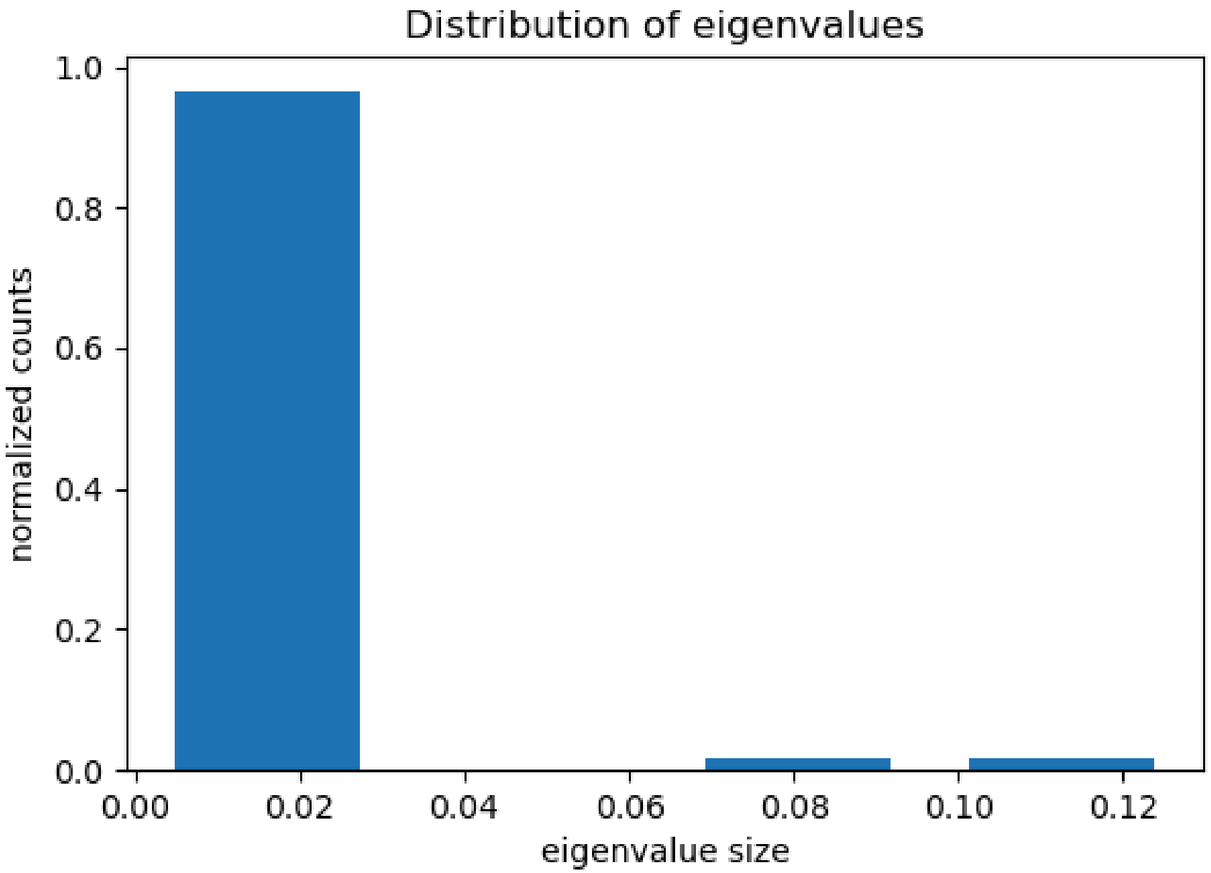
Barren plateaus

Are regions of the parameter space where: $|\frac{1}{N}\sum_i \mathbb{E}_\theta[\frac{d}{d\theta_j} log(L(x_i, y_i, \theta)]| < c$

This implies: $tr(F(\theta)) \leq d(c + c^2)$

Using Fisher information we can understand if the model is affected by the Barren Plateuas phenomenon

Fisher spectrum for classical model



PCA

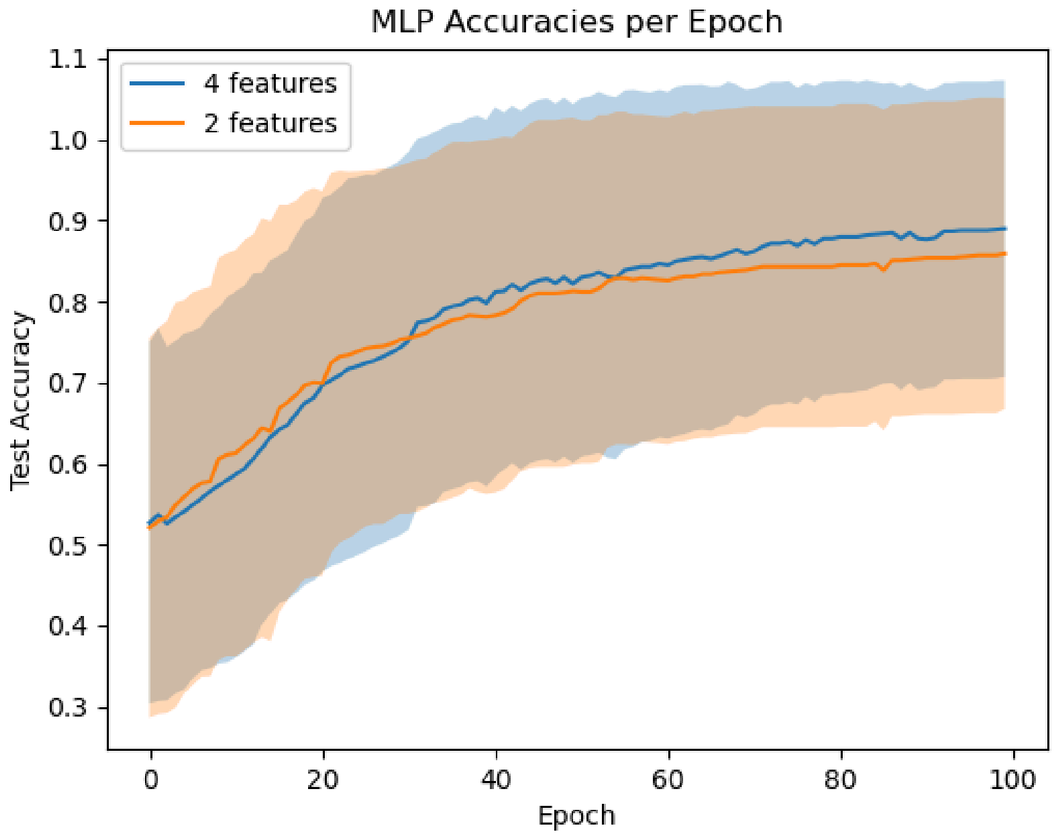
$$X_{nsamples,4} \longrightarrow X_{nsamples,2}$$

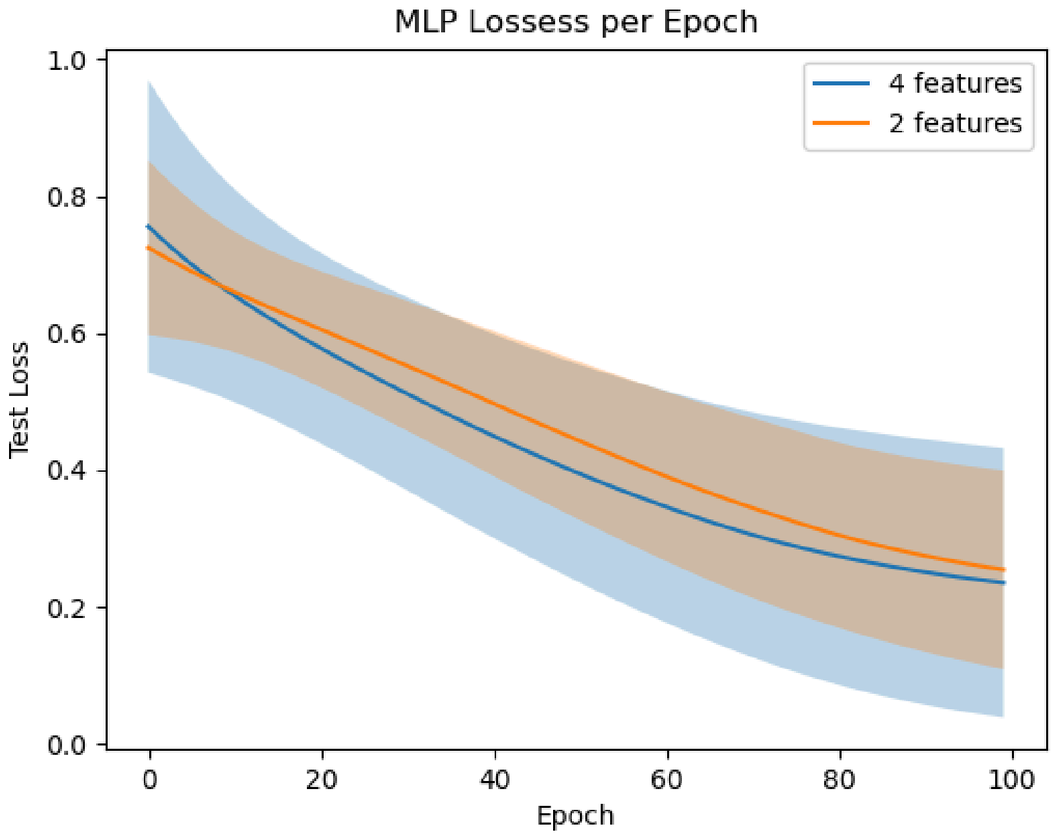
Trough a principal component analysis
of the covariance matrix:

$$C = X^T X$$

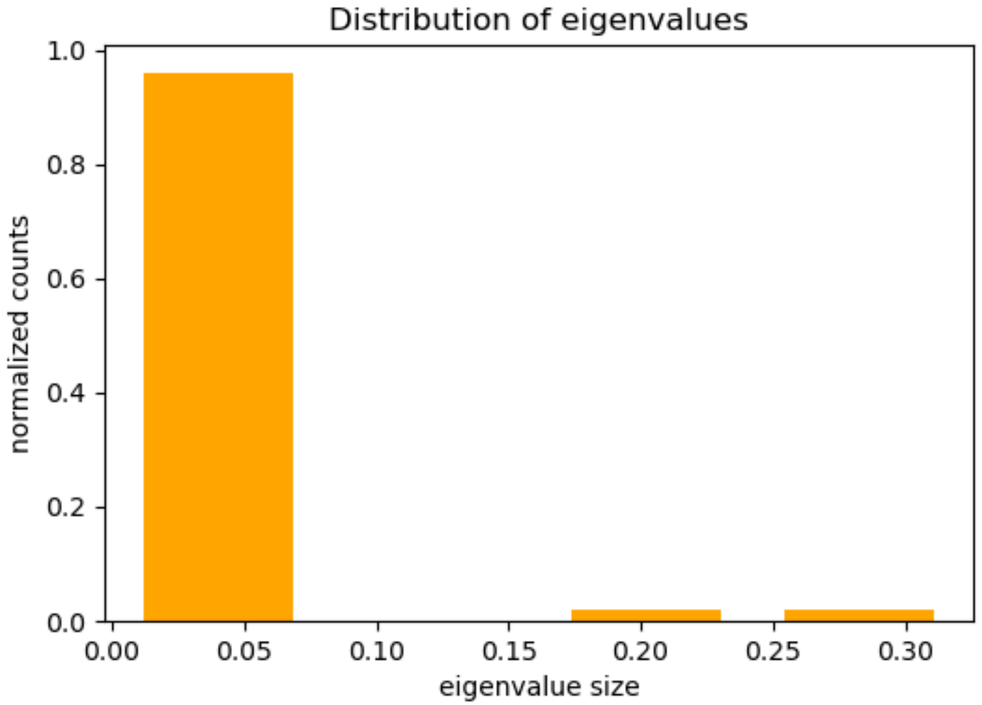
**The covariance matrix is correlated
to the effectie dimension**

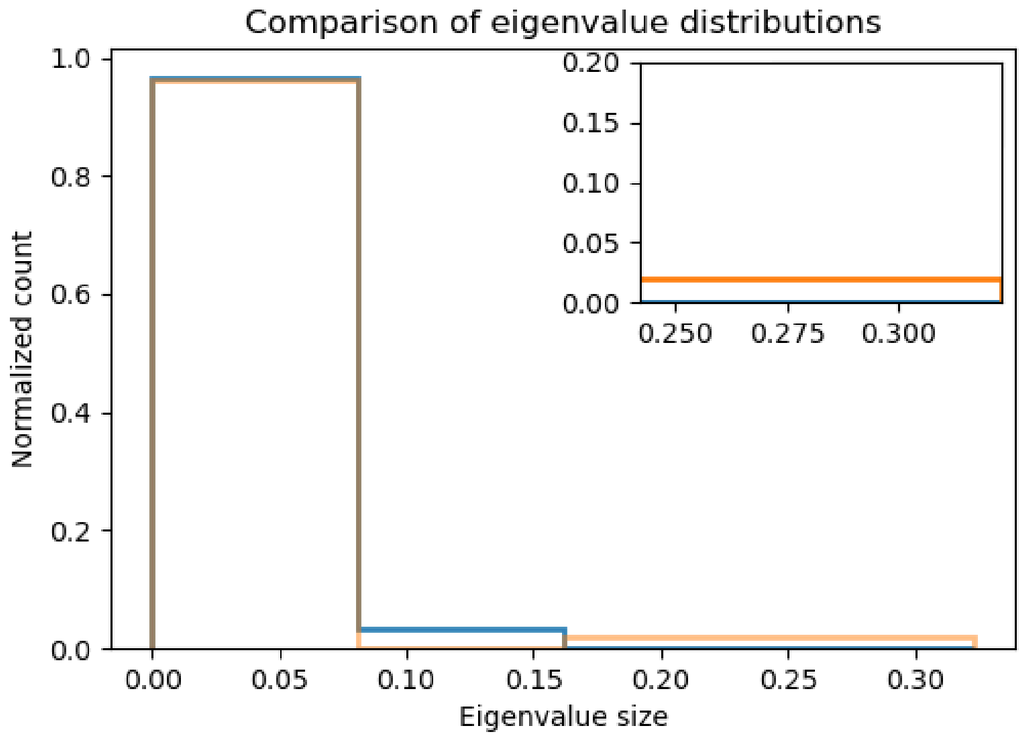
Performances of dataset after PCA



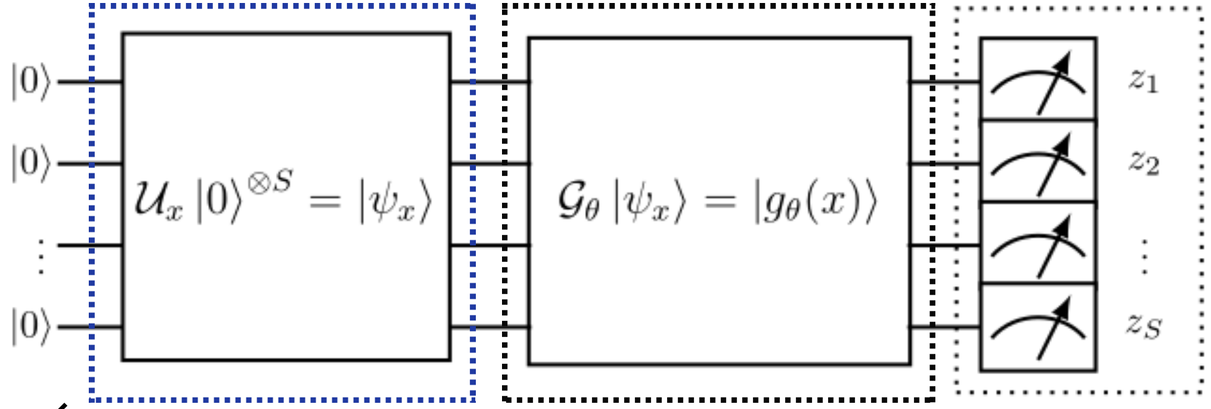


Distribution of eigenvalues after PCA

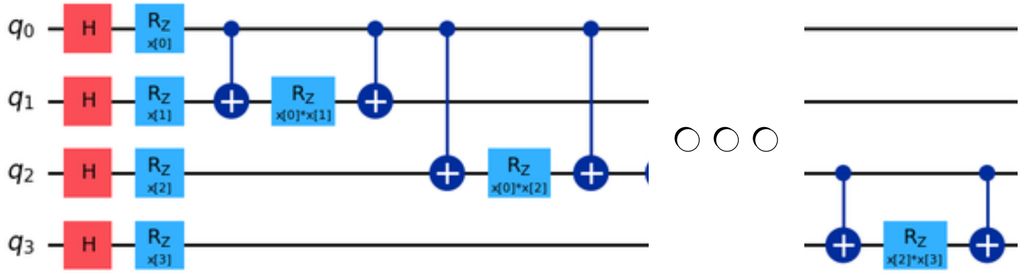


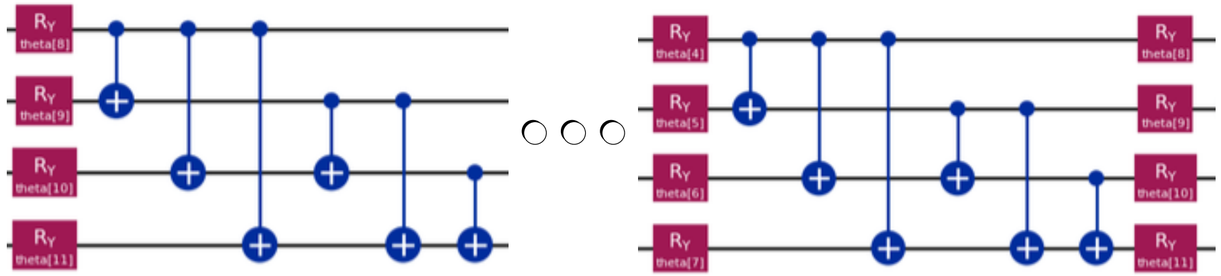


Quantum model

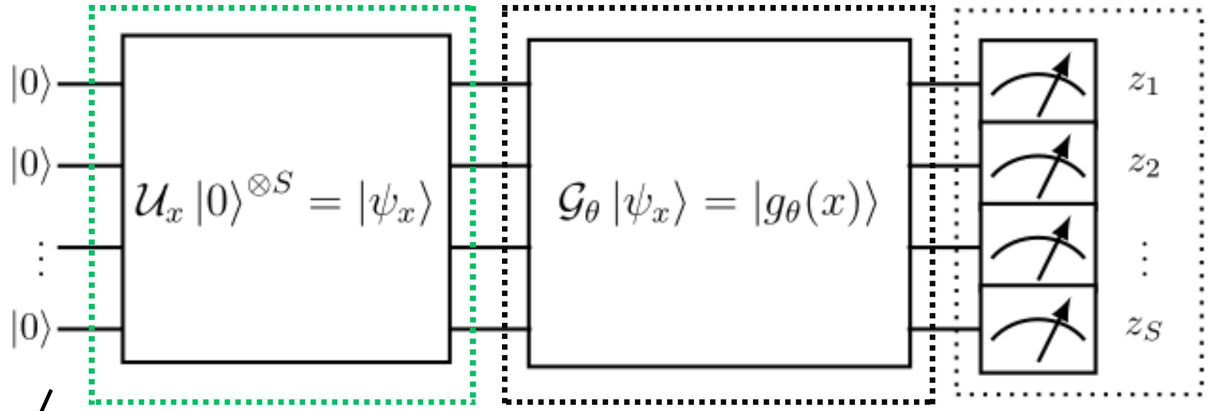


- 4 qubits
- 60 parameters
- depth 111
- 2-qubit gates: 102

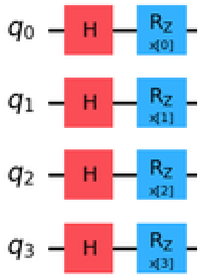


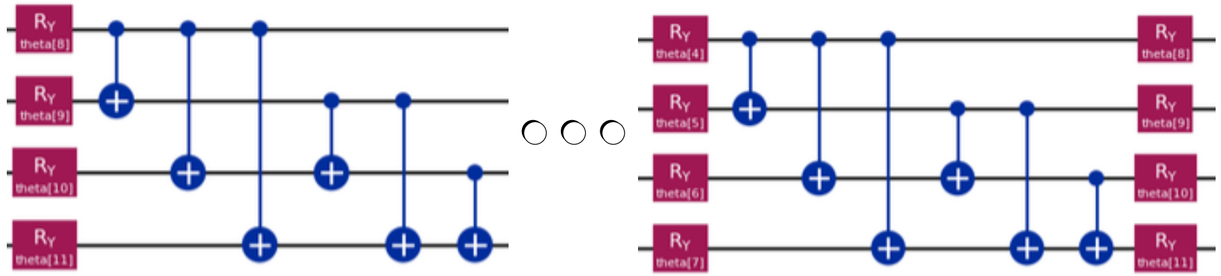


Easy Quantum model



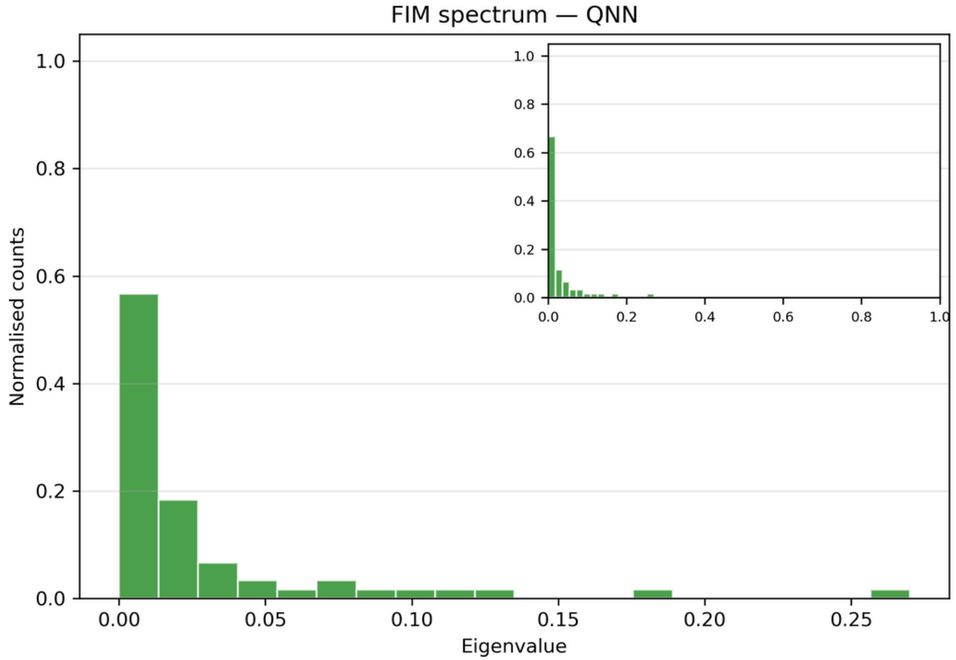
- 4 qubits
- 60 parameters
- depth 93
- 2-qubit gates: 90

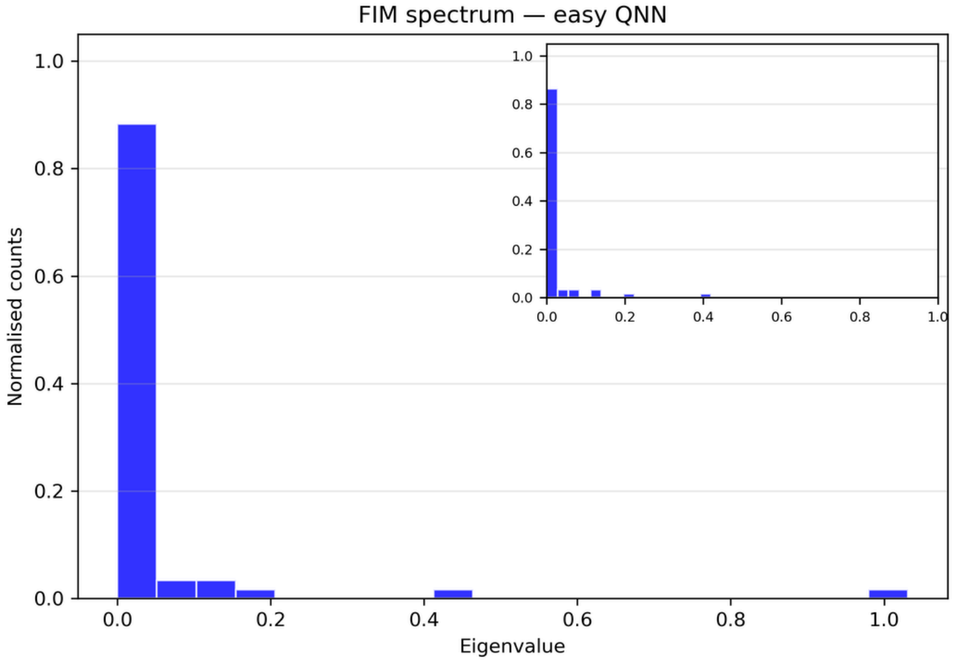


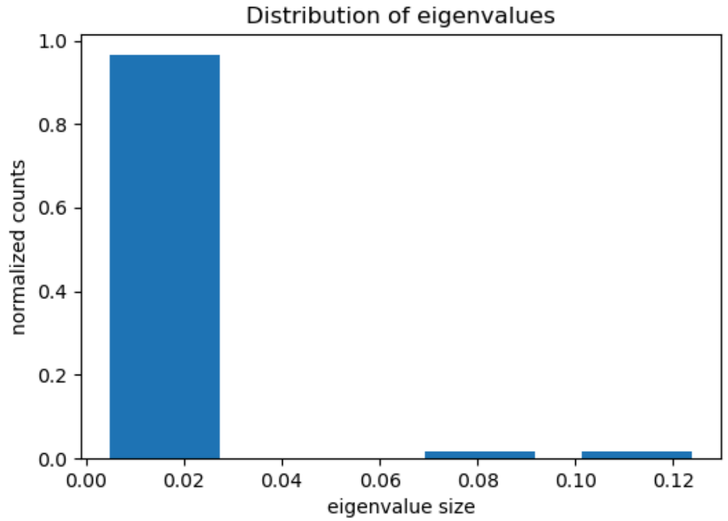


For both models L function is cross-entropy

Fisher Information for QNN



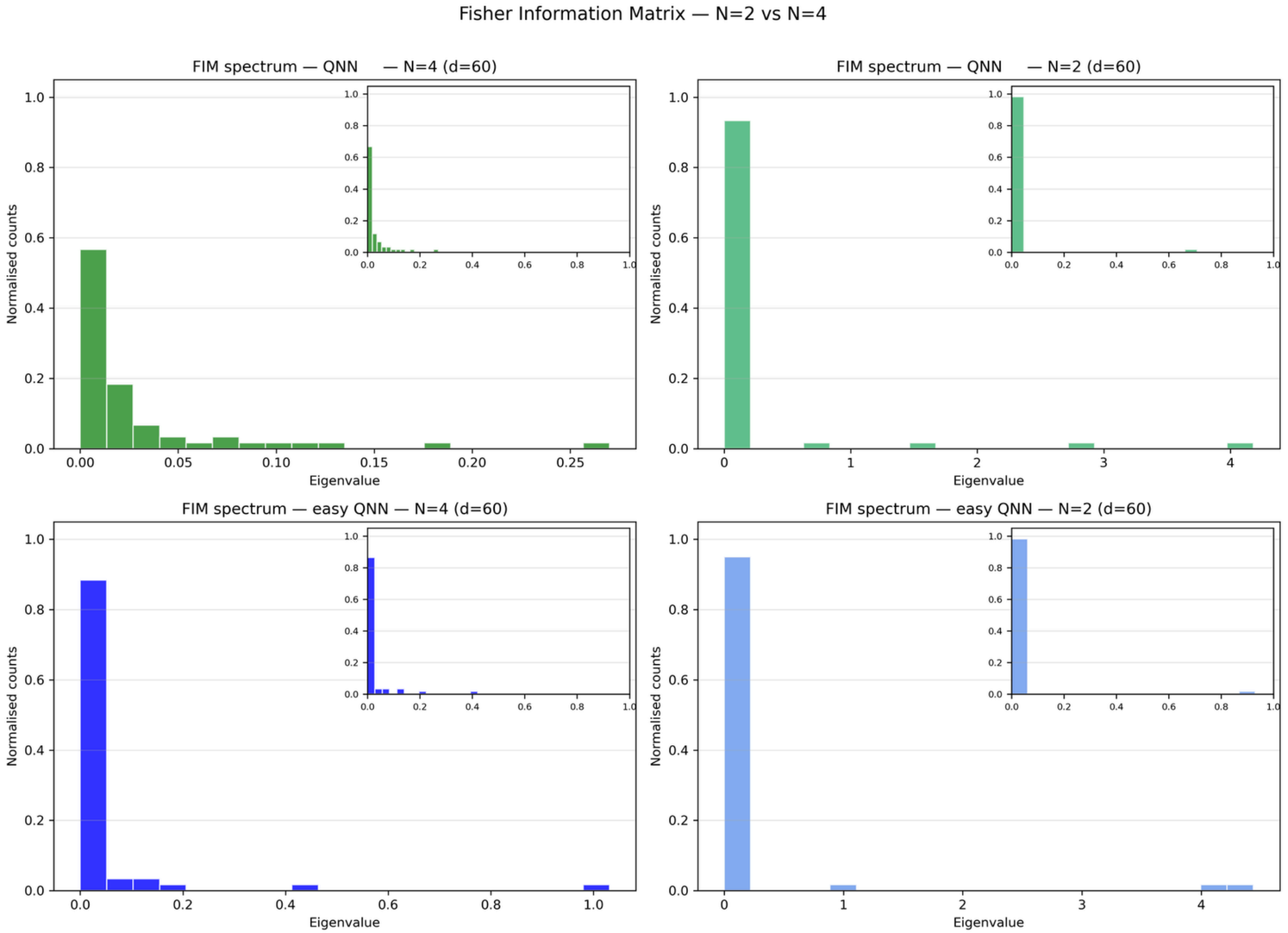




QNNs have better FI distribution (w.r.t. Classical NN)

More resilient against barren plateus

Confrontation with 2 qubits



FI eigenvalues with 2 qubits is more concentrated in 0: More susceptible to barren plateus

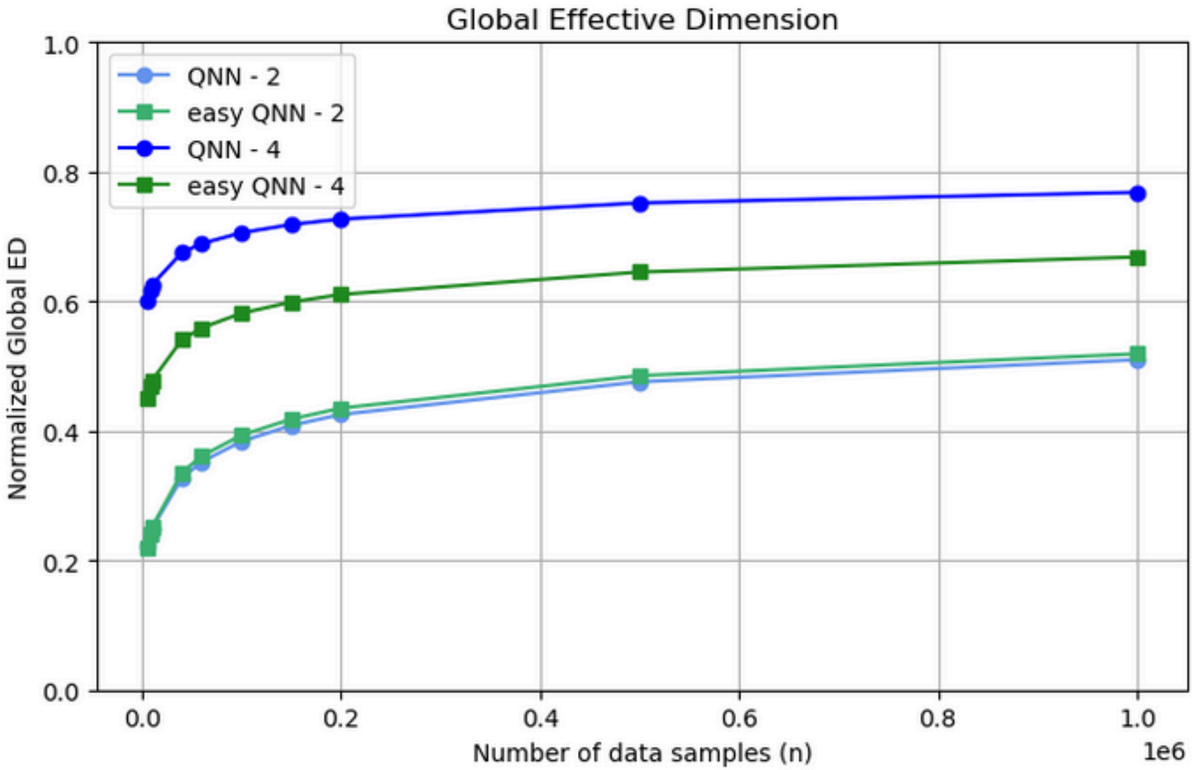
Effective Dimension

Definition: $$\dim_{\mathrm{eff},n}(\mathcal{M}) := 2\frac{\log\left(\frac{1}{V_\Theta}\int_\Theta \sqrt{\det\left(\mathrm{Id}_d + \frac{n}{2\pi}\,\hat{\boldsymbol{F}}(\theta)\right)}\,d\theta\right)}{\log\frac{n}{2\pi}}$$

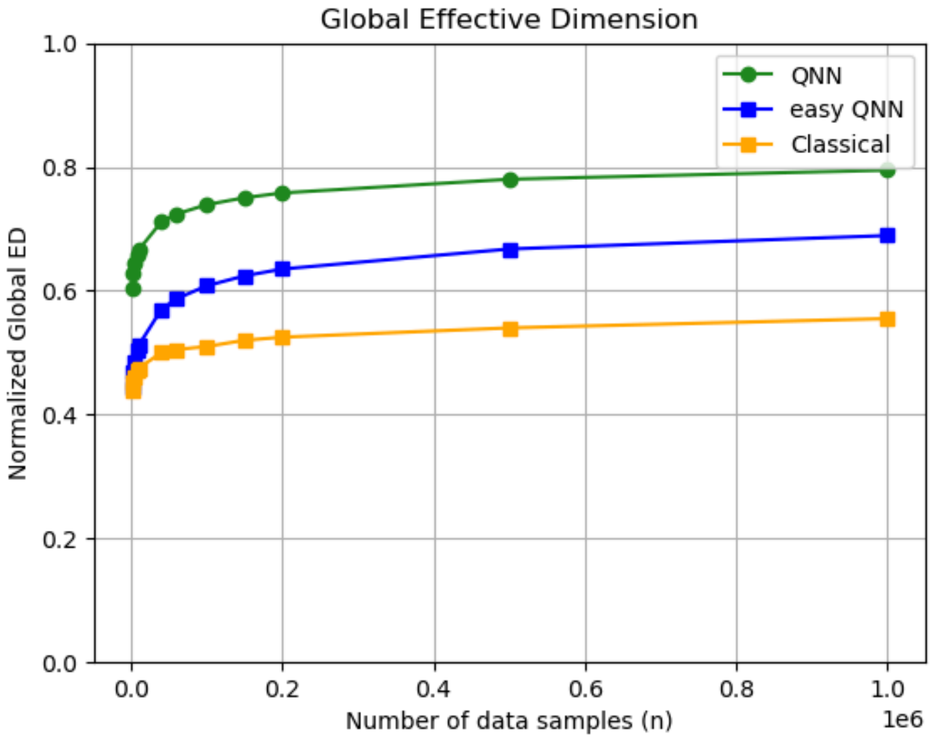
It “estimates size occupied by a model in the model space” (Abbas et all)
Expression of the ability of a model to cover of all its possible functions

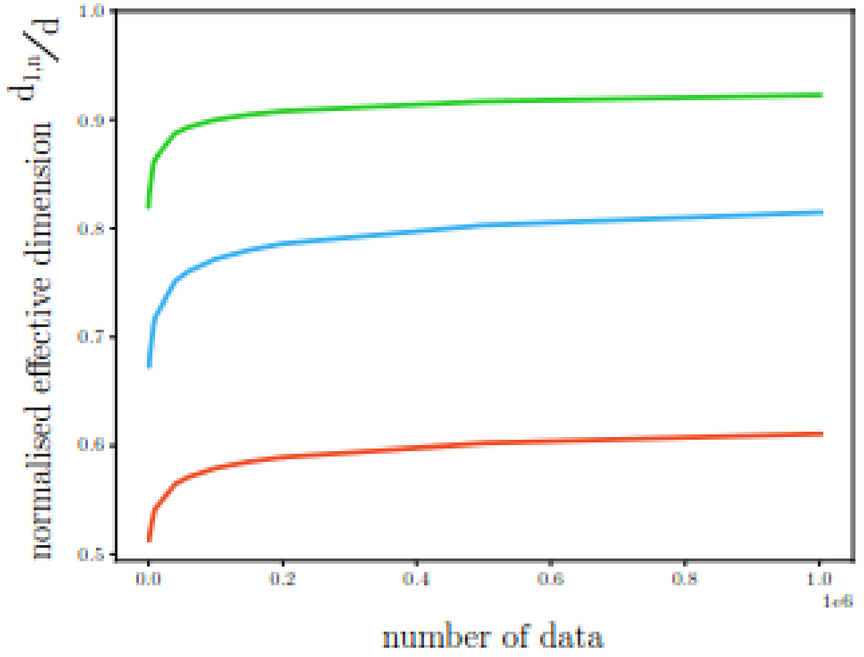
- Dependant on Fisher Information: $$\hat{F}_{ij}(\theta) := d\frac{V_\Theta}{\int_\Theta \mathrm{tr}(F(\theta))\mathrm{d}\theta}F_{ij}(\theta)$$
- Asymptotically $\dim_{\mathrm{eff},n}(\mathcal{M}) \to d \quad \text{as } n \to \infty.$

ED for 2-4 qubits



ED of all models





QNNs are more expressive in the model space than Classical

Digression on ED & FI

What happens when all FI eigenvalues are identical?

Assume $F = c\mathbb{1}, \hat{F} = \mathbb{1}$ $k = \frac{n}{2\pi}$

Then

$$\sqrt{det(\mathbb{1} + k\mathbb{1})} = (1+k)^{\frac{d}{2}}$$

$$\dim_{\mathrm{eff},n}(\mathcal{M}) := 2\frac{\log\left(\frac{1}{V_\Theta}\int_\Theta \sqrt{\det\left(\mathrm{Id}_d + \frac{n}{2\pi}\hat{\boldsymbol{F}}(\theta)\right)}\, d\theta\right)}{\log\frac{n}{2\pi}}$$

$$ED = 2\frac{log(1+k)^{\frac{d}{2}}}{log(k)} = 2\frac{d}{2}\frac{log(1+k)}{log(k)} = d\frac{log(1+k)}{log(k)} \approx 1$$ for “low” n! (1000: 1.01)

Still, it’s not feasible in reality

Also, if $c \to 0$ we encounter barren plateus: trainability is an issue

Capacity analysis

Global vs local effective dimension

GLOBAL ED

$$d_{\gamma,n}(\mathcal{M}_\Theta) := 2\frac{\log\left(\frac{1}{V_\Theta}\int_\Theta \sqrt{\det\left(\mathrm{id}_d + \frac{\gamma n}{2\pi\log n}\hat{F}(\theta)\right)}\,\mathrm{d}\theta\right)}{\log\left(\frac{\gamma n}{2\pi\log n}\right)}$$

It’s computed over the whole parameter space

LOCAL ED

$$\theta^\star = \arg\min_\theta \mathcal{L}(\theta)$$

$$d_{\mathrm{local}}(n,\theta^\star) = \frac{1}{\log\left(\frac{n}{2\pi\log n}\right)}\log\det\left(I + \frac{n}{2\pi\log n}\bar{F}(\theta^\star)\right)$$

$$\bar{F}(\theta^\star) = \frac{d\,F(\theta^\star)}{\mathrm{Tr}(F(\theta^\star))}$$

It’s computed on a single set of weights, those obtained from the optimization of the model (trained weights)

Capacity analysis
Global vs local effective dimension

GLOBAL ED

It's a measure of the model's **architectural expressivity**: *how rich the family of functions the model could represent*

The generalisation error is bounded by an expression which depends on the global effective dimension ("The power of quantum neural networks")

$$\mathbb{P}\left(\sup_{\theta\in\Theta}|R(\theta)-R_n(\theta)|\geq 4M\sqrt{\frac{2\pi\log n}{\gamma n}}\right)\leq c_{d,\Lambda}\left(\frac{\gamma n^{1/\alpha}}{2\pi\log n^{1/\alpha}}\right)^{\frac{a_{\gamma,n^{1/\alpha}}}{2}}\exp\left(-\frac{16M^2\pi\log n}{B^2\gamma}\right)$$

LOCAL ED

It measures the **complexity** of the model **after training**: *it represents the number of parameter directions in which the model is meaningfully sensitive around the trained solution* θ^*.

- **HIGH LOCAL ED**: small parameters changes shift the output a lot → **poor generalization** (risk of overfitting)
- **LOW LOCAL ED**: the model has "settled" → **good generalization**

("Effective dimension of machine learning models")

Capacity analysis
Fisher information matrix

In the computation of both the global and the local effective dimension we use the Fisher information matrix. For a single input x and set of weights θ, the model outputs a probability distribution over C classes. The true Fisher information at that input is defined as:

$$F(x,\theta) = \sum_{c=1}^{C} p_\theta(c \mid x)\, \nabla_\theta \log p_\theta(c \mid x)\, \big(\nabla_\theta \log p_\theta(c \mid x)\big)^\top$$

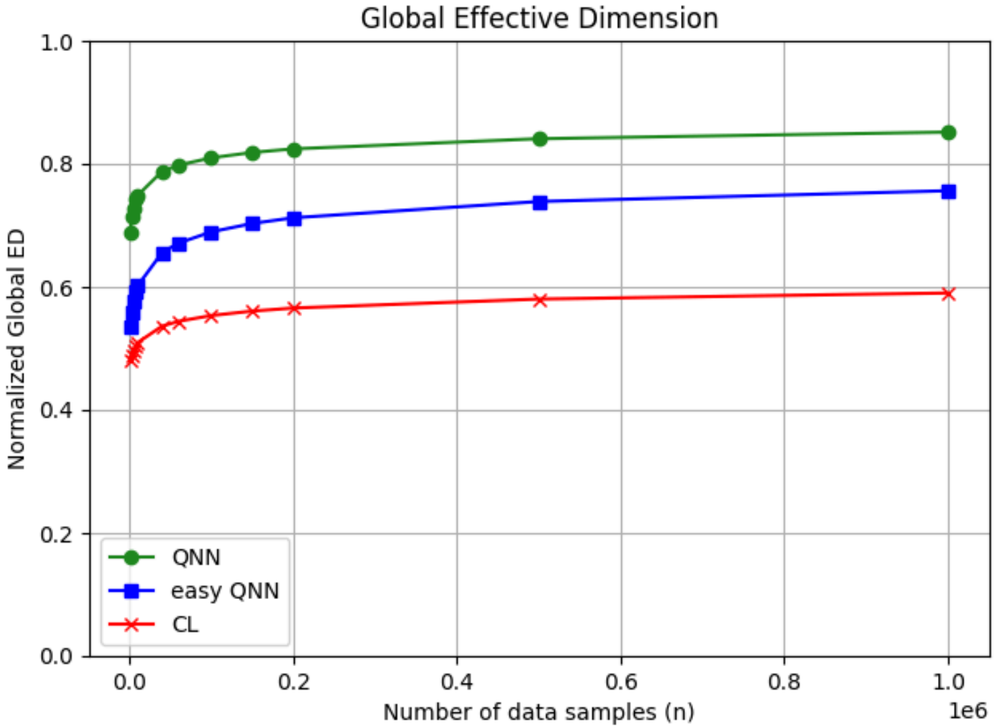
Then we compute the average over all input samples in the dataset:

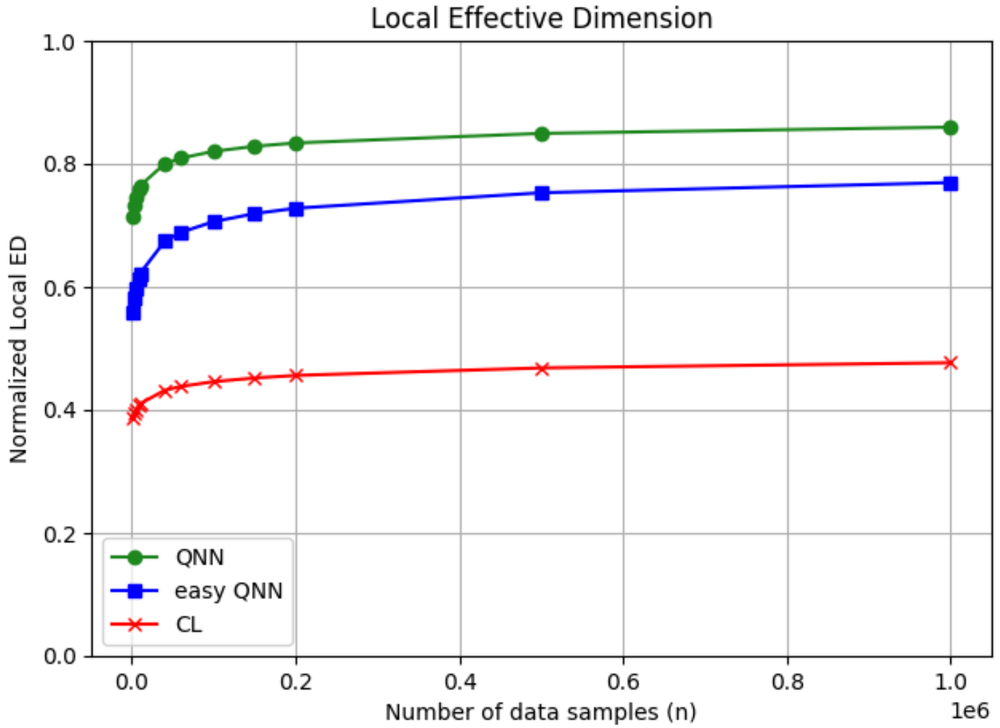
$$F(\theta) = \frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} p_\theta(c \mid x_i)\, \nabla_\theta \log p_\theta(c \mid x_i)\, \big(\nabla_\theta \log p_\theta(c \mid x_i)\big)^\top$$

Capacity analysis

Global vs local effective dimension

SYNTHETIC DATASET

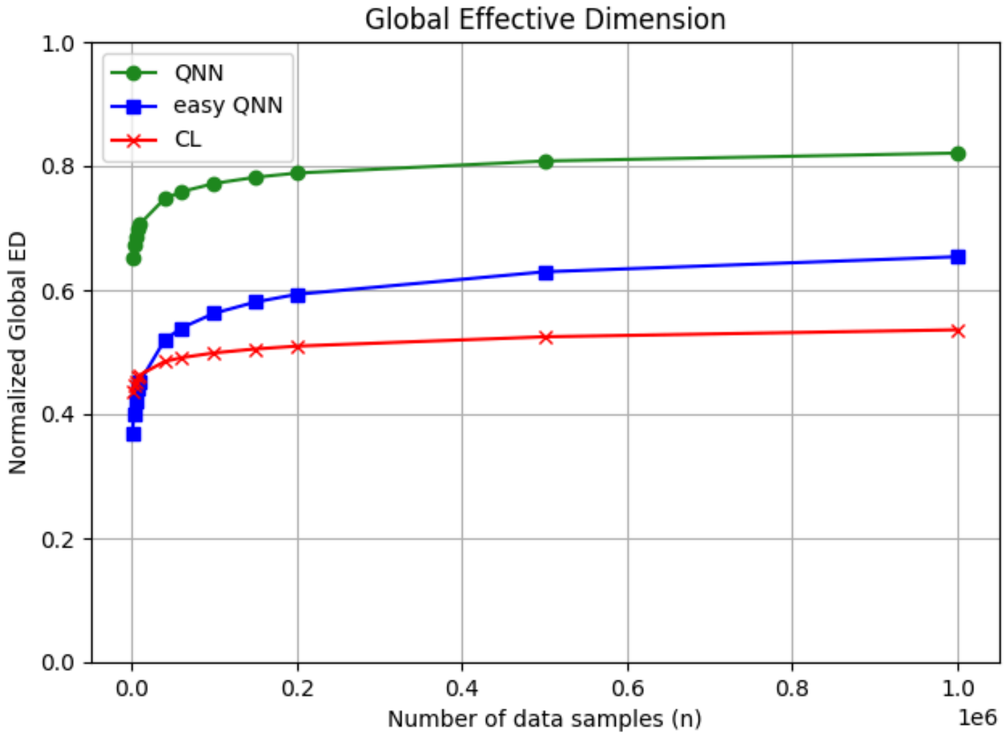


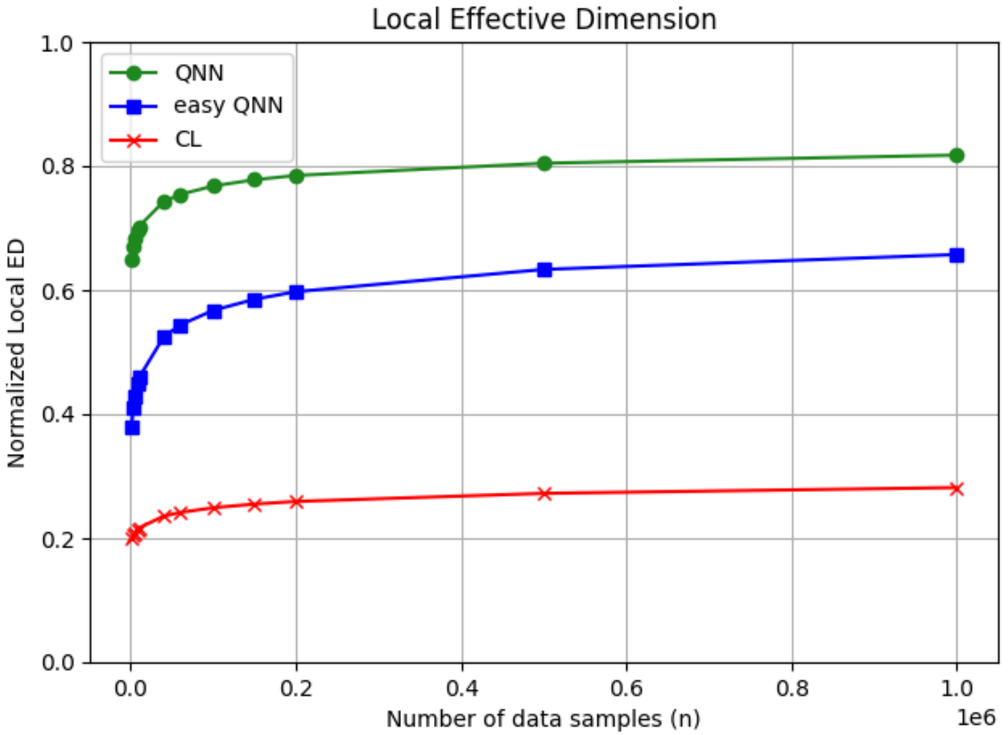


Capacity analysis

Global vs local effective dimension

BINARY DATASET

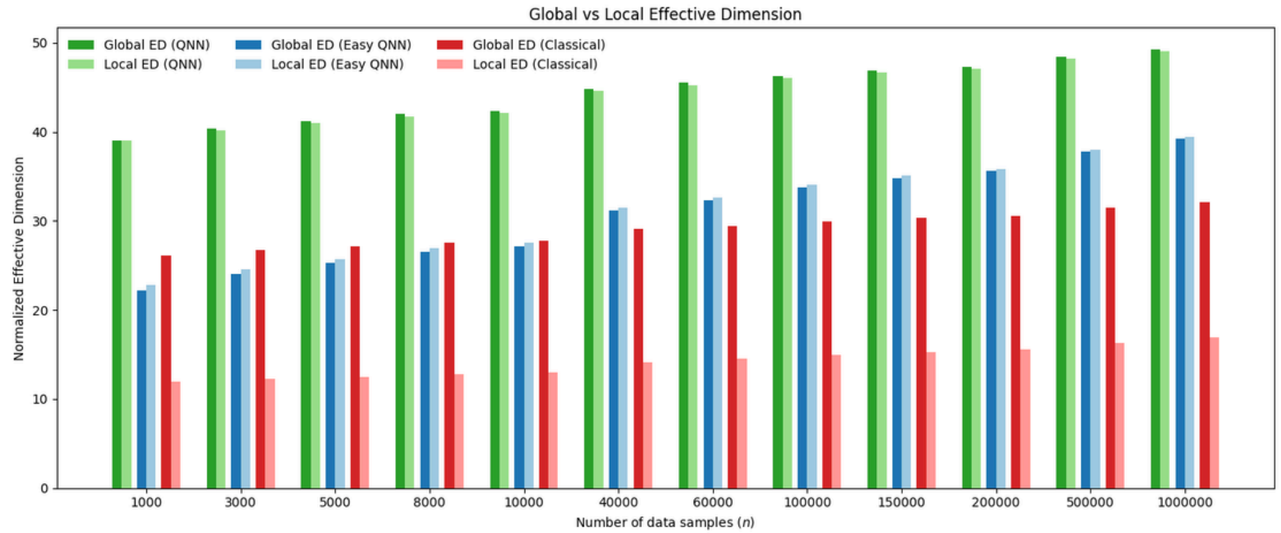




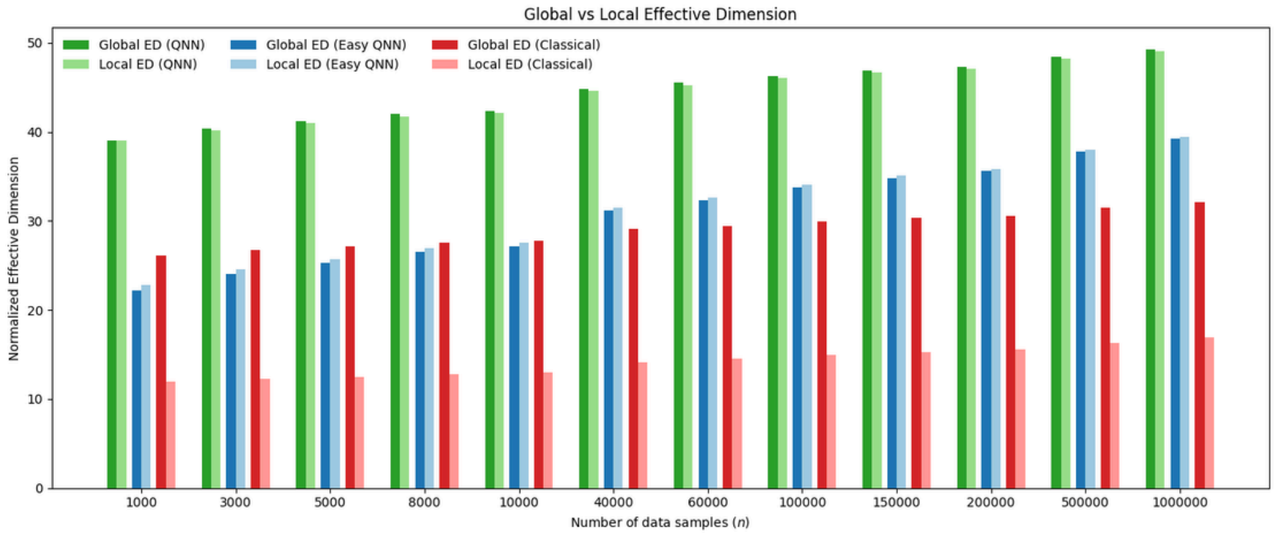
Capacity analysis

Global vs local effective dimension

SYNTHETIC DATASET

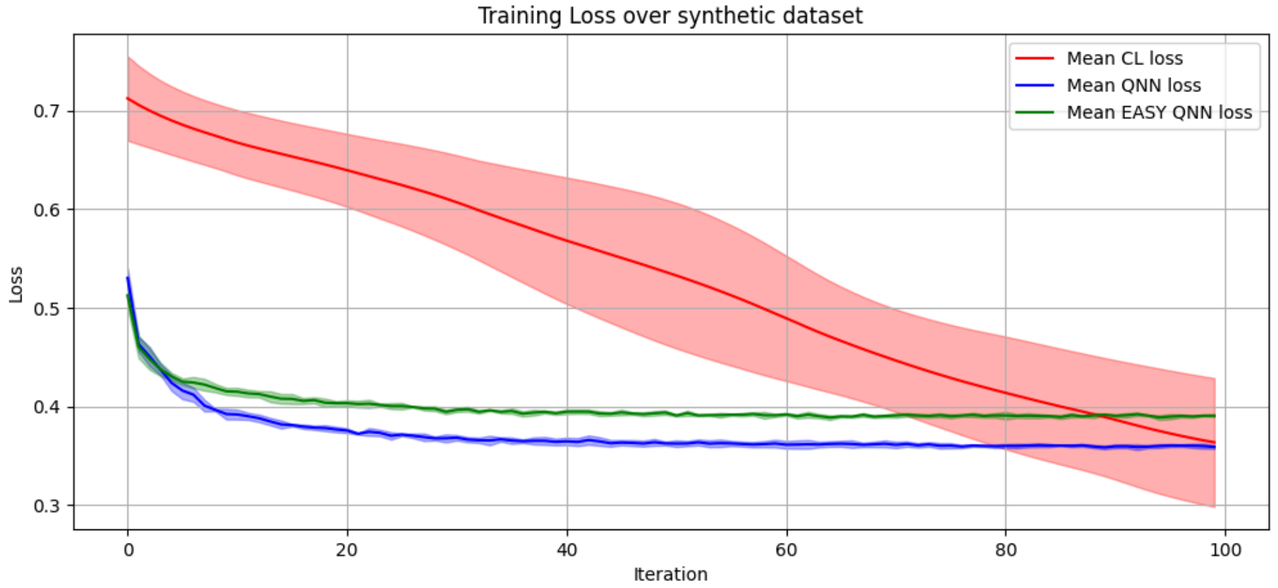


BINARY DATASET



Capacity analysis
Training loss

SYNTHETIC DATASET

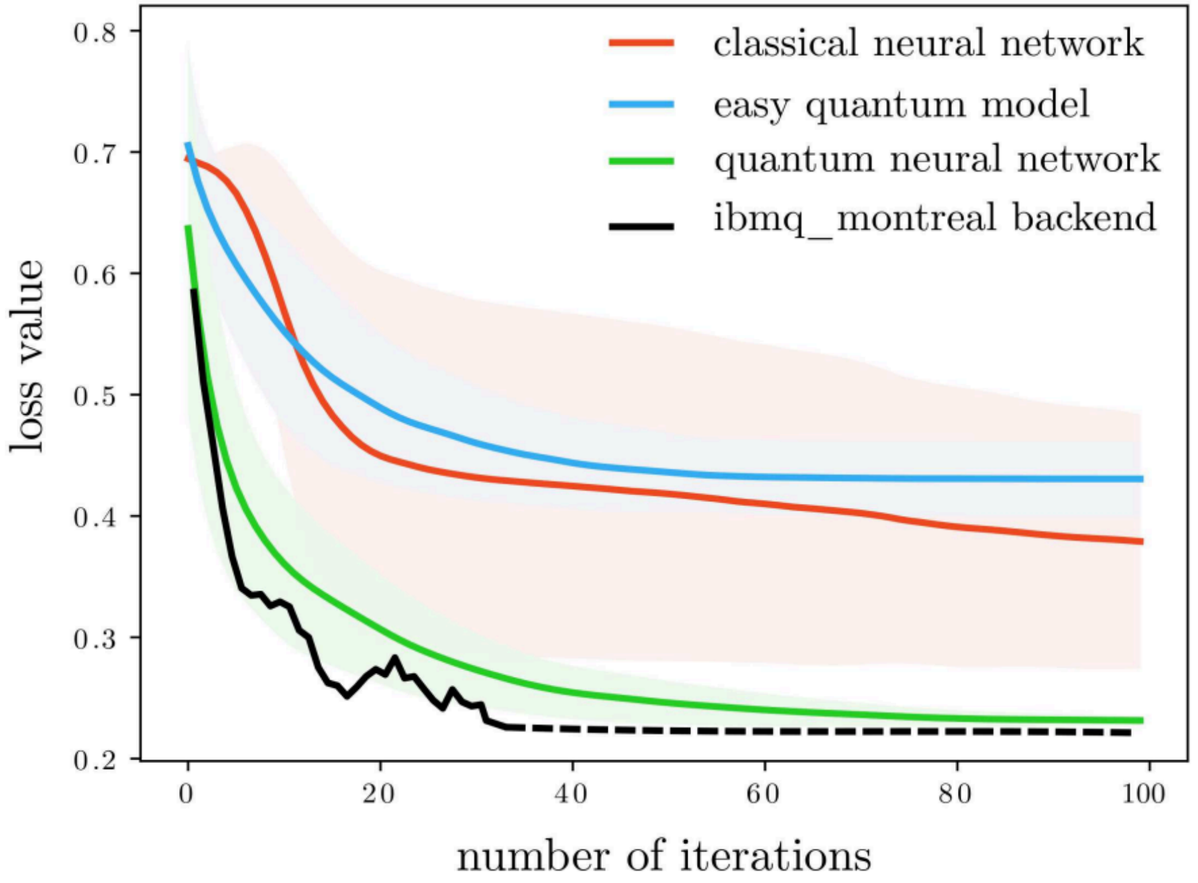


BINARY DATASET



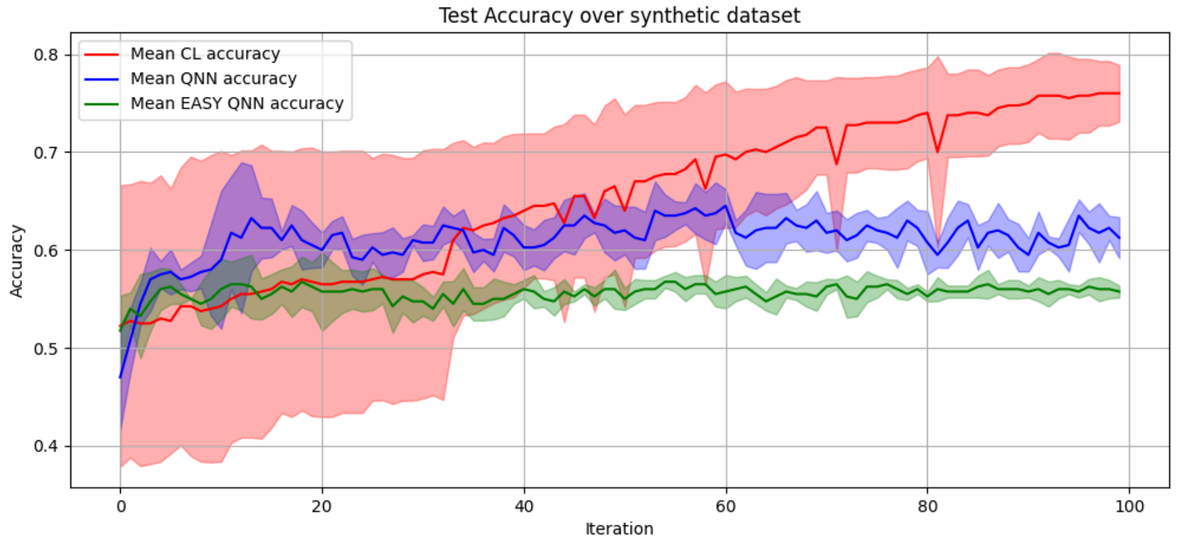
Capacity analysis

Training loss - comparison with the paper's results

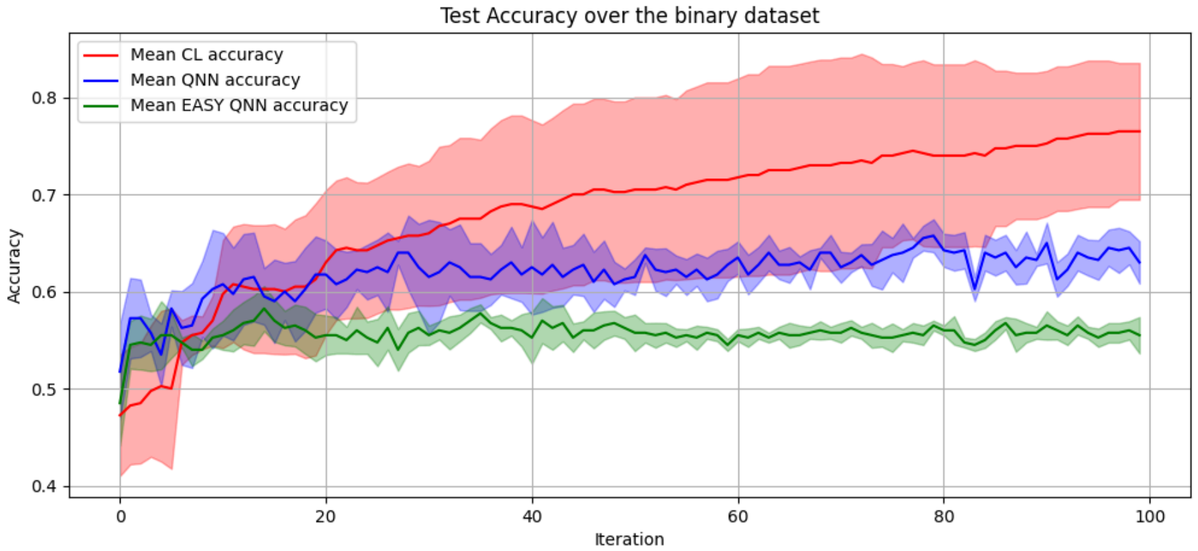


Capacity analysis
Test accuracy

SYNTHETIC DATASET

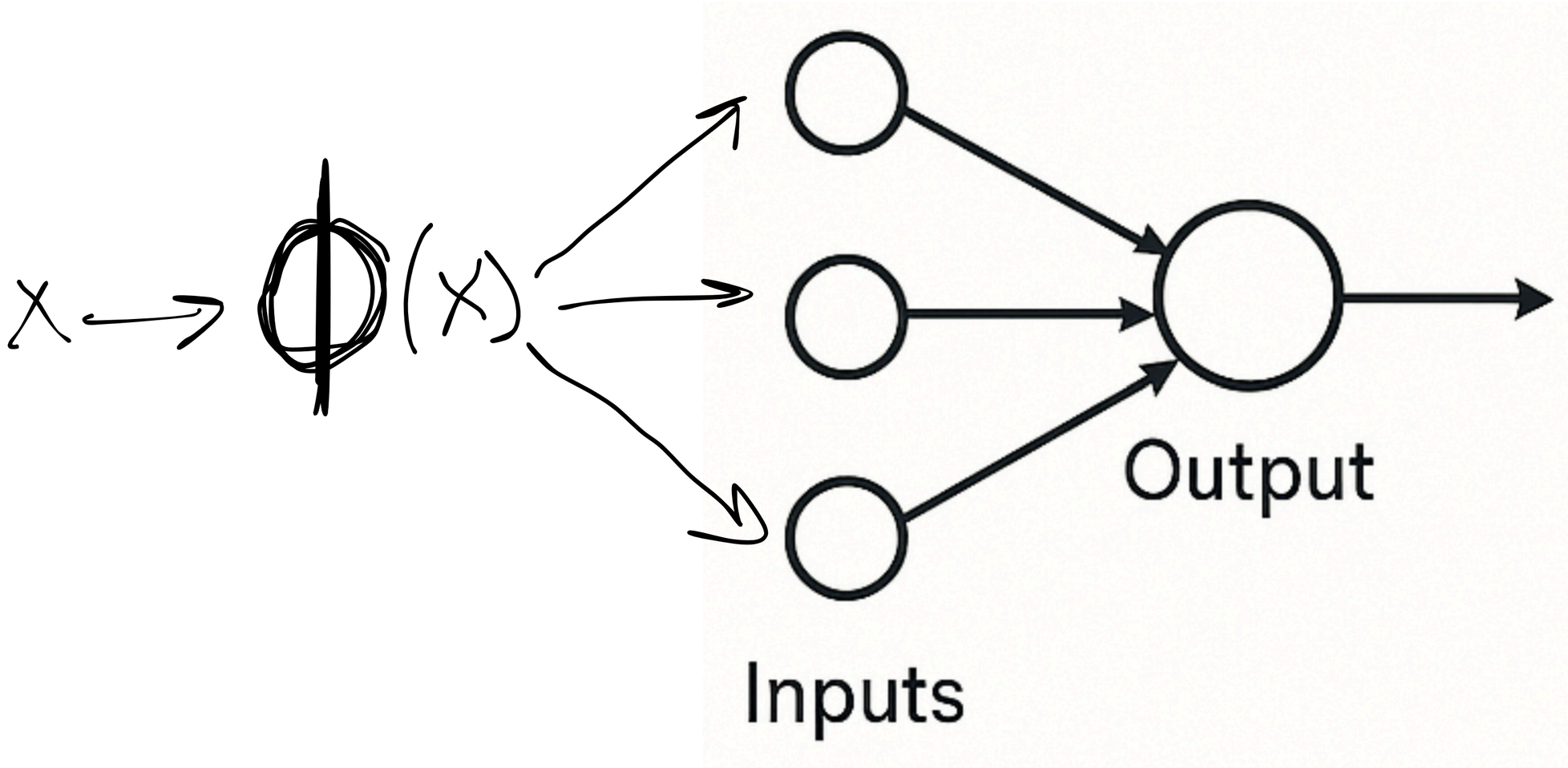


BINARY DATASET



Dividing feature map from final neuron

Neural Networks representation:



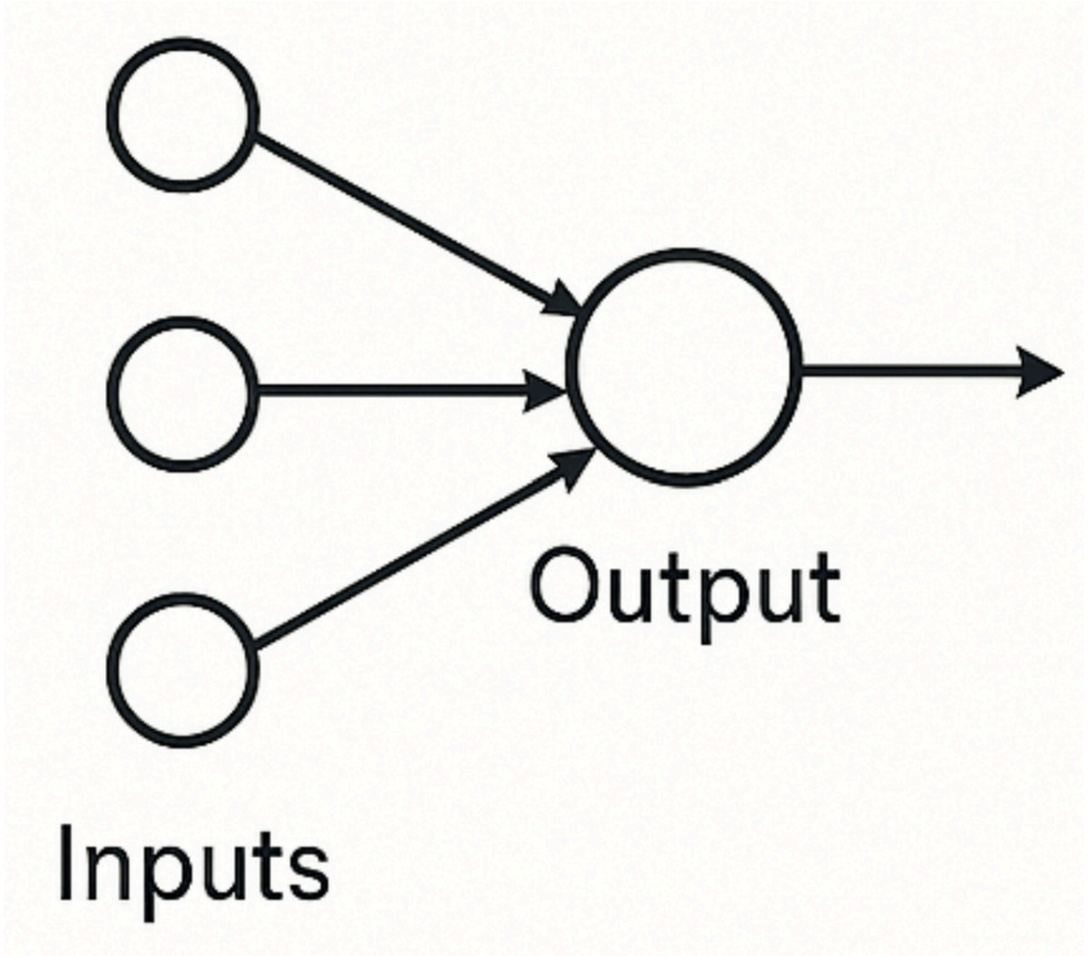
What we want to do?

- Divide the last neuron (the classifier/regressor) from the feature map
  - Doing that we can understand if the quantum neuron alone is more, less or equally powerful if compared to a classical neuron
  - We can understand the same for the feature map as well

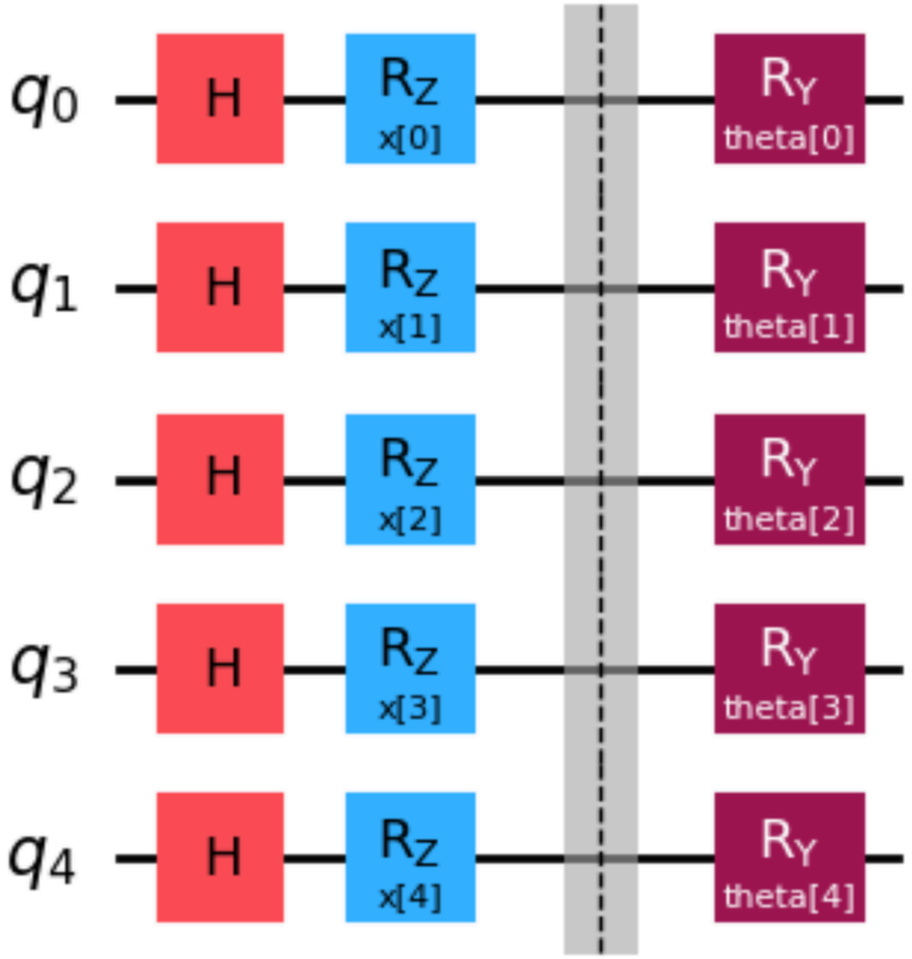
Last neuron analysis

Models

Classical



Quantum



Experiment’s Problem Selection

I. Variance around x=0

$$\hat{x} = \frac{x}{x_{max} + x_{min}} \qquad \tilde{x} = \cos^{-1}(\hat{x})$$

$$H|0\rangle = |+\rangle_x \qquad R_z(\tilde{x})|+\rangle = |\psi\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\tilde{x}}|1\rangle\right)$$

$$R_y(\theta)|\psi\rangle = \frac{1}{\sqrt{2}}\left((\cos\frac{\theta}{2} - e^{i\tilde{x}}\sin\frac{\theta}{2})|0\rangle + (\cos\frac{\theta}{2} + e^{i\tilde{x}}\sin\frac{\theta}{2})|1\rangle\right)$$

$$p_0 = \frac{1 - \cos\tilde{x}\sin\theta}{2} \qquad p_1 = \frac{1 + \cos\tilde{x}\sin\theta}{2}$$

Experiment’s Problem Selection

$$p(k) = \frac{n!}{(n-k)!k!} p_0^k p_1^{n-k}$$

$$f(\hat{x}) = 2(\frac{k}{n} - \frac{1}{2})$$

$$\langle f \rangle (\hat{x}) = 2(\frac{\langle k \rangle}{n} - \frac{1}{2}) = 2(\frac{np_0}{n} - \frac{1}{2}) = -\cos\tilde{x}\sin\theta = \hat{x}\sin\theta$$

$$Var(\langle f \rangle) = \frac{4}{n^2} Var(k) = \frac{4}{n^2} np_0p_1 = \frac{1-\cos\tilde{x}\sin\theta}{n} = \frac{1-\hat{x}\sin\theta}{n}$$

Experiment’s Problem Selection

No generalization outside the min-max interval!

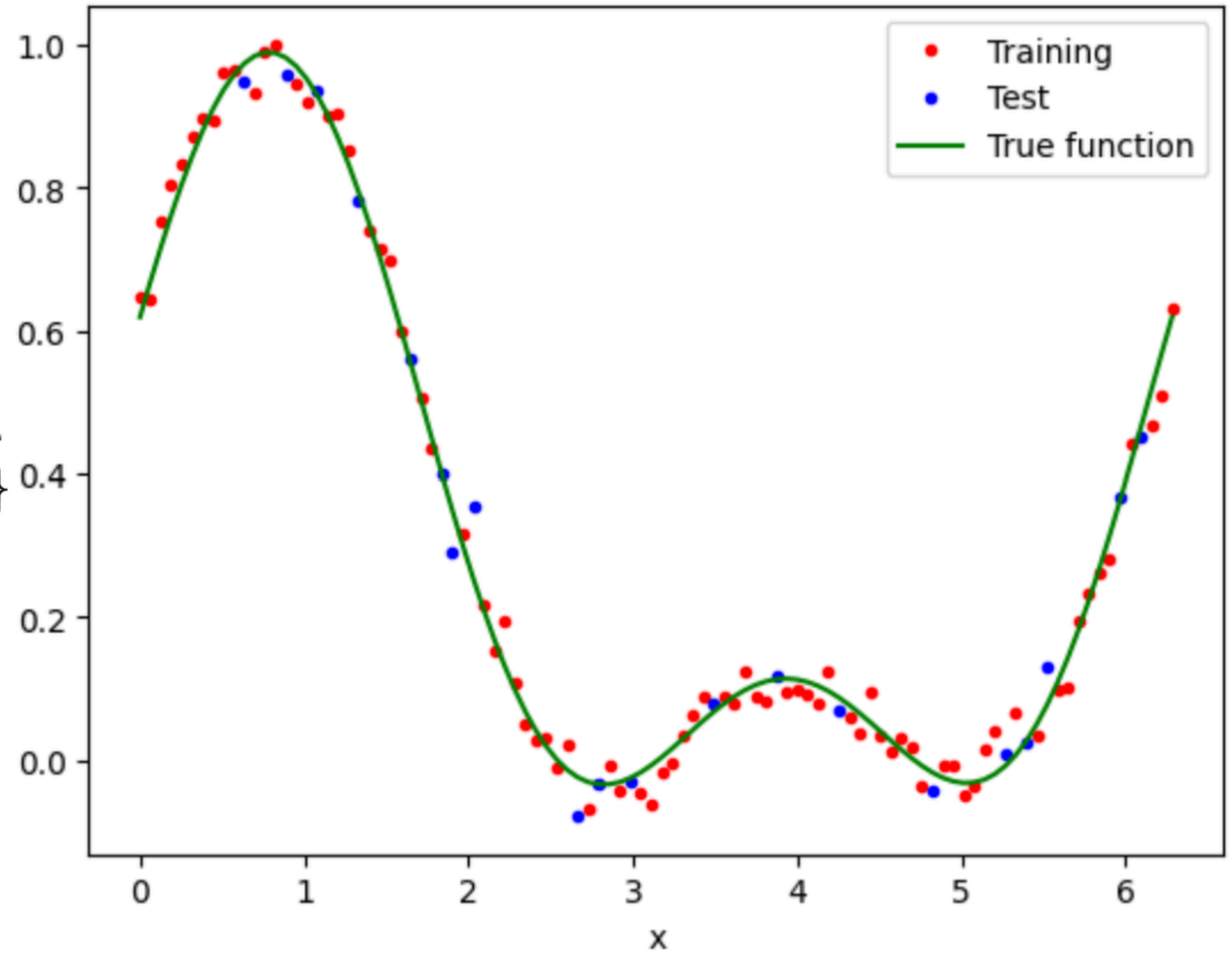
Experiment’s Problem Selection

Sinusoidal model:

- Do not needs scaling
- Easy state preparation for a sinusoidal feature map

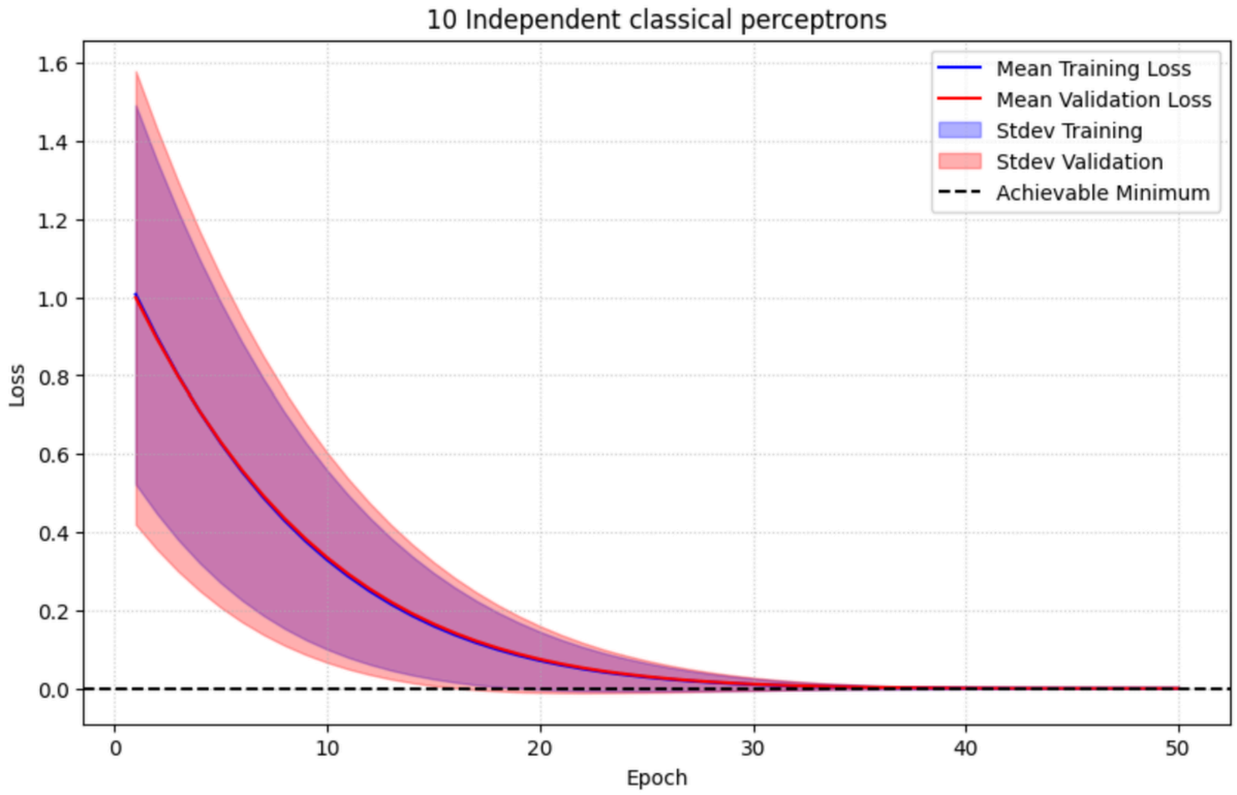
$$\phi(x) = \{1, \cos x, \sin x, \cos 2x, \sin 2x\}$$

$$\phi(x) = \{1, \cos x, \cos(x - \frac{\pi}{2}), \cos 2x, \cos(2x - \frac{\pi}{2}))\}$$

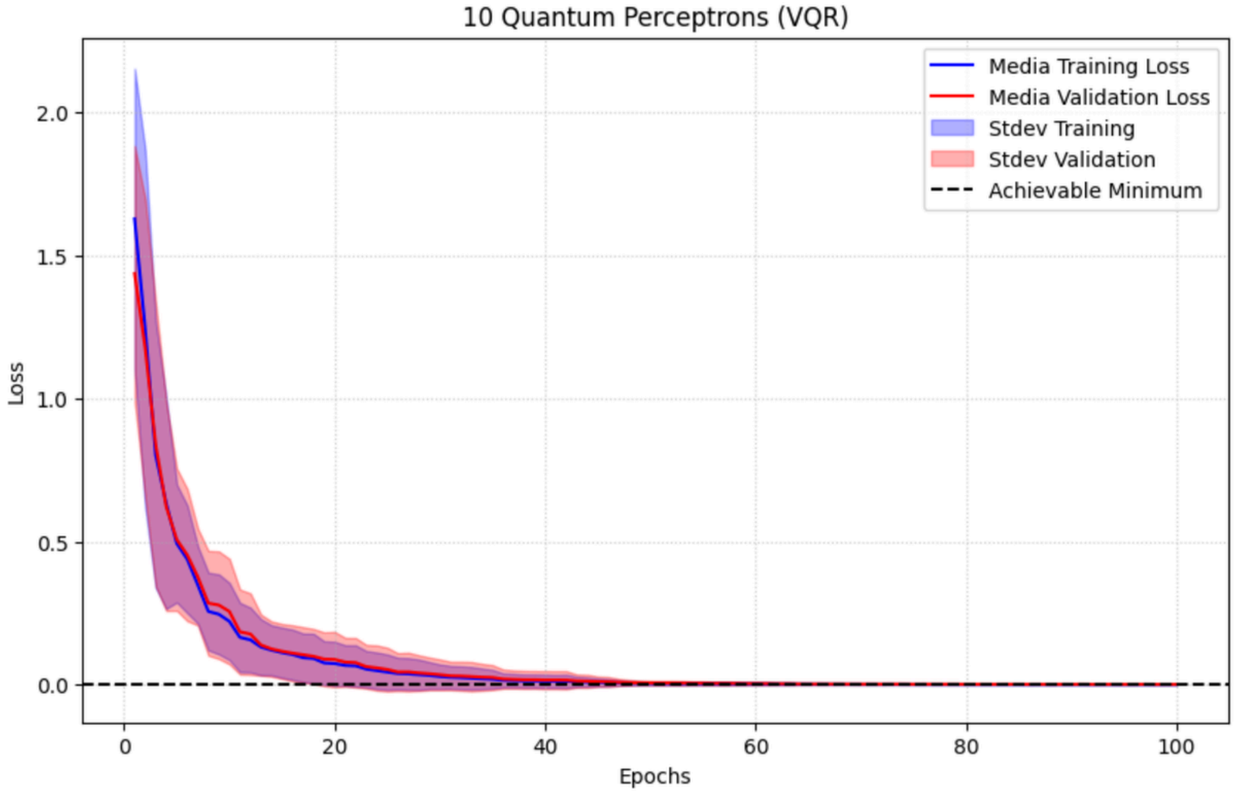


Loss

Classical

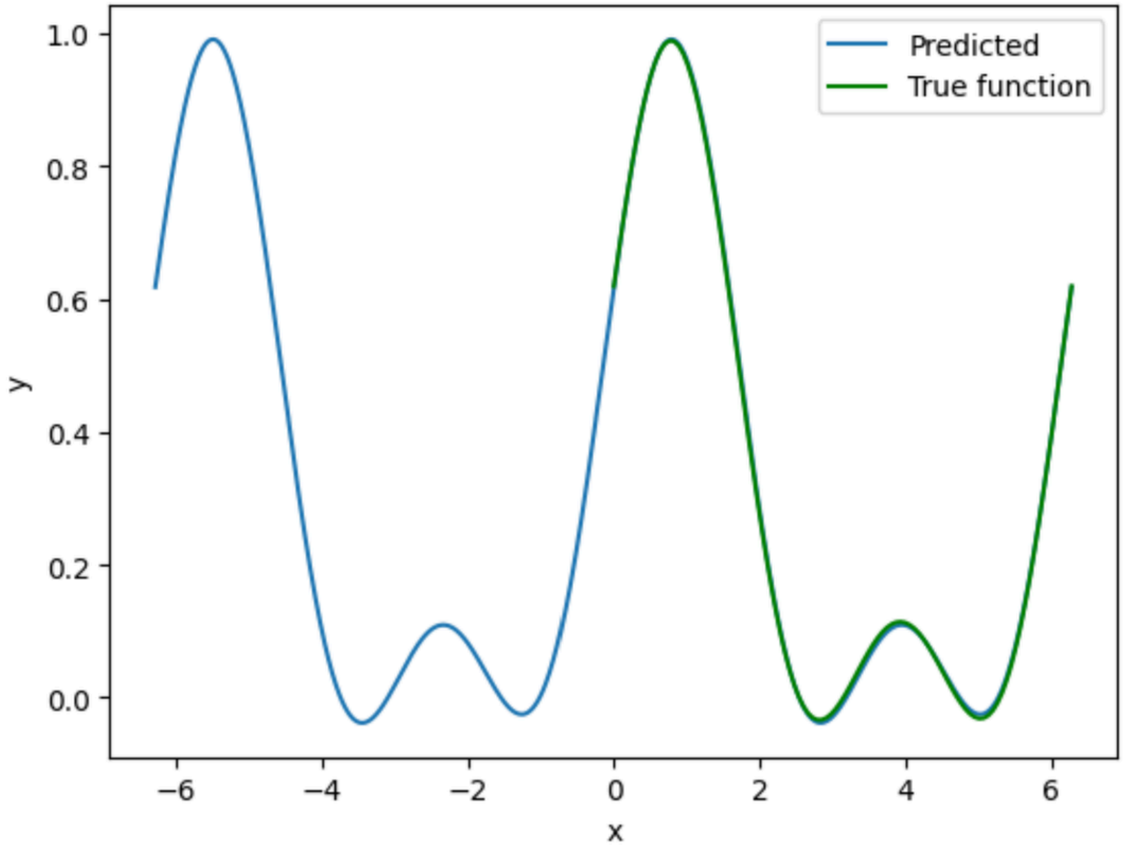


Quantum

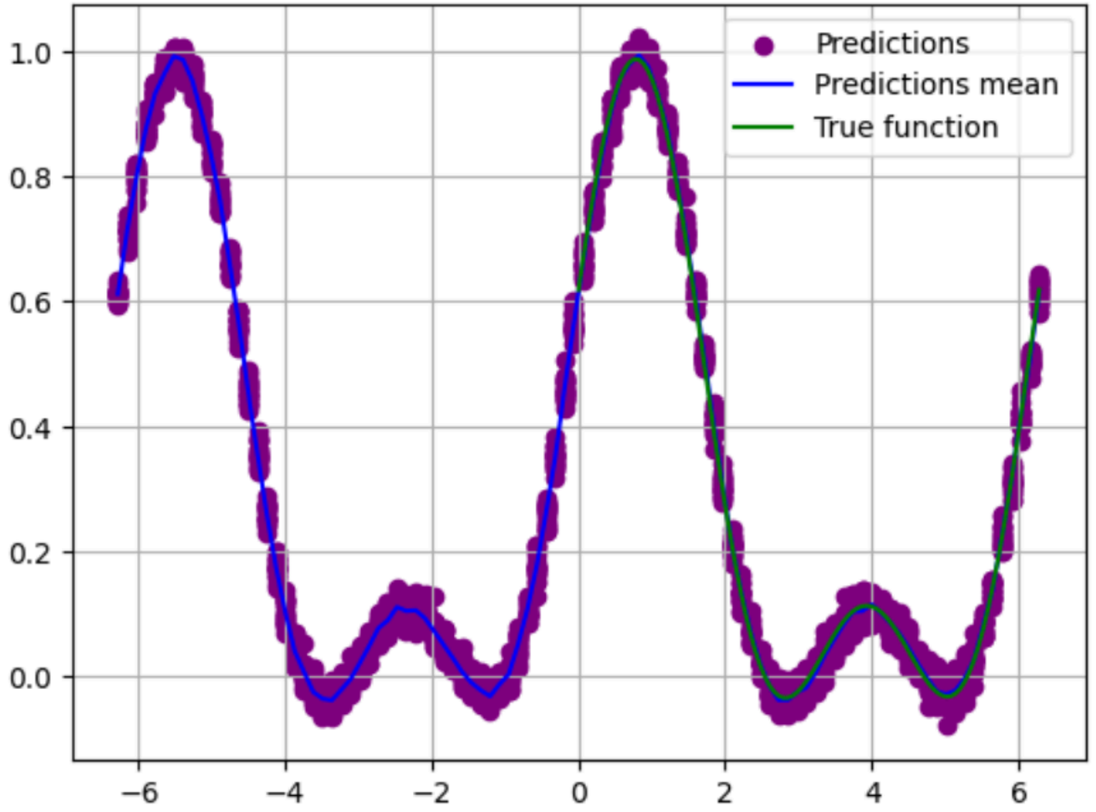


Prediction

Classical



Quantum



Effective dimension

Why the classical neuron seems to have the higer power?

$$I^{classic} = \frac{1}{\sigma^2}\mathbf{x}\mathbf{x}^\top, \qquad I^{quantum} = \frac{1}{\sigma_{eff}^2}\mathbf{u}\mathbf{u}^\top + \frac{1}{2\sigma_{eff}^4}\mathbf{v}\mathbf{v}^\top$$

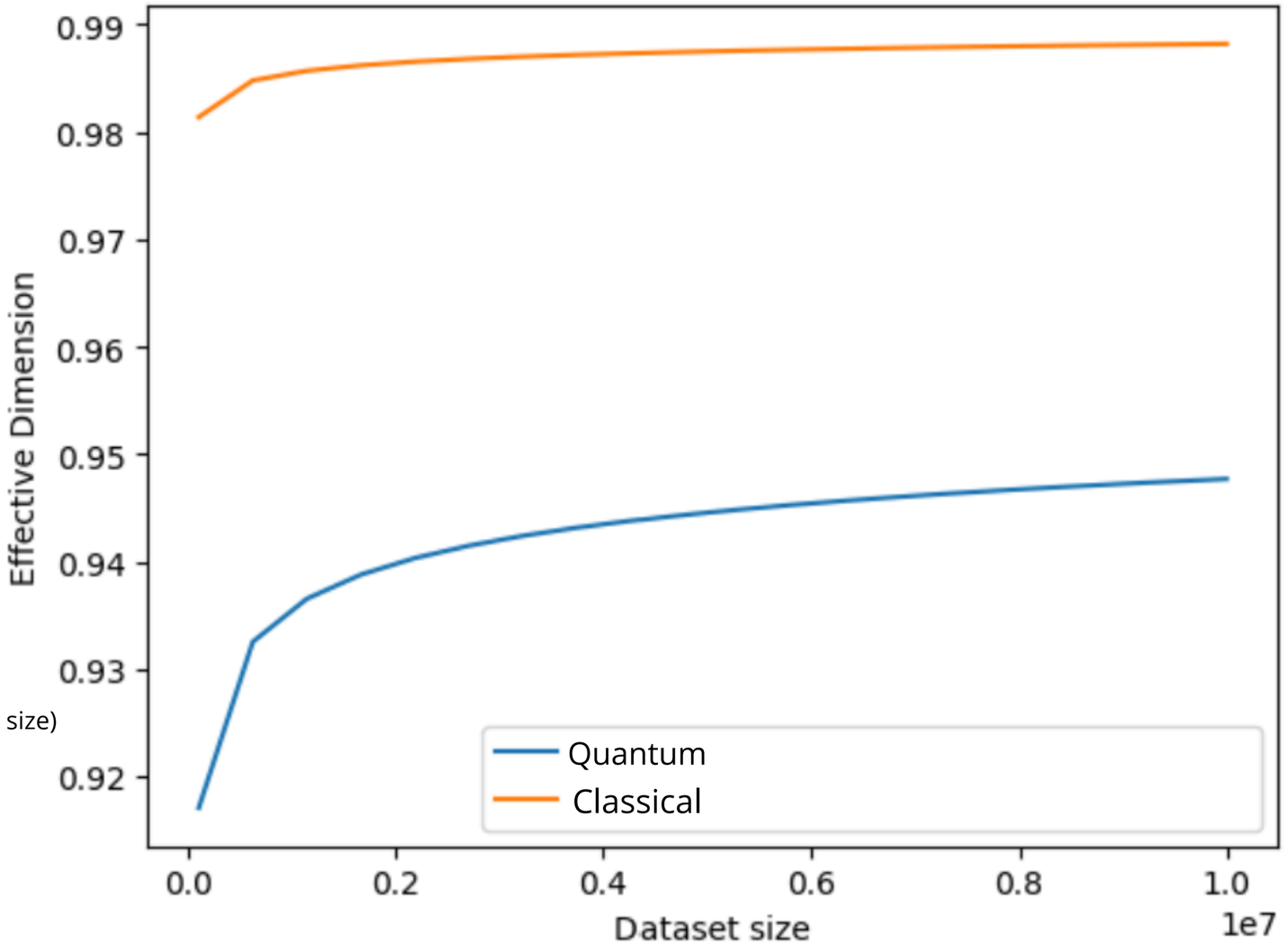
$$u_i = \hat{x}_i \cos\theta_i, \qquad v_i = \frac{2\hat{x}_i^2 \sin\theta_i \cos\theta_i}{n}$$

$$I^{classic} = \frac{1}{\sigma^2}\mathbf{x}\mathbf{x}^\top, \qquad I^{quantum} = \frac{1}{\sigma_y^2}\mathbf{u}\mathbf{u}^\top$$

$$d_{\gamma,n} = \bar{r} \qquad n \to \infty$$ (Here n is the dataset size)

Where $\bar{r} = \max_{\theta\in\Theta} Rank(I(\theta))$

Asimptotically the same!



So is the quantum neuron more powerful?

General Vapnik–Chervonenkis dimension bounds for quantum circuit learning

Chih-Chieh Chen[1,*], Masaru Sogabe[1], Kodai Shiba[1,2], Katsuyoshi Sakamoto[2,3] and Tomah Sogabe[1,2,3,*]

Theorem 1 (VC dimension upper bound for quantum circuits). *Assume the input vector* $\vec{x} = (x_0, \ldots, x_{d-1}) \in [-1,1]^d$. *Each dimension* $x_i \in [-1,1]$ *is uploaded* n_i *times using single qubit encoding rotations* $R_s(\phi_i(x_i))$ *for some fixed encoding mapping* $\phi_i : [-1,1] \to [-\pi,\pi]$ *with* $s \in \{Y,Z\}$. *Then the VC dimension of the hypothesis set* $H = \{\mathrm{sgn}(f_\theta(\vec{\phi}(\vec{x})) + c) : f_\theta(\vec{\phi}(\vec{x})) = \langle O(\theta, \vec{\phi}(\vec{x}))\rangle = \mathrm{Tr}[OU_\theta(\vec{\phi}(\vec{x}))|0\rangle^{\otimes n} \langle 0|^{\otimes n} U_\theta^\dagger(\vec{\phi}(\vec{x}))], c \in \mathbb{R}\}$ *for a fixed observable* O *is upper bounded by:*

$$d_{VC} \leqslant \prod_{i=0}^{d-1} (2n_i + 1).$$

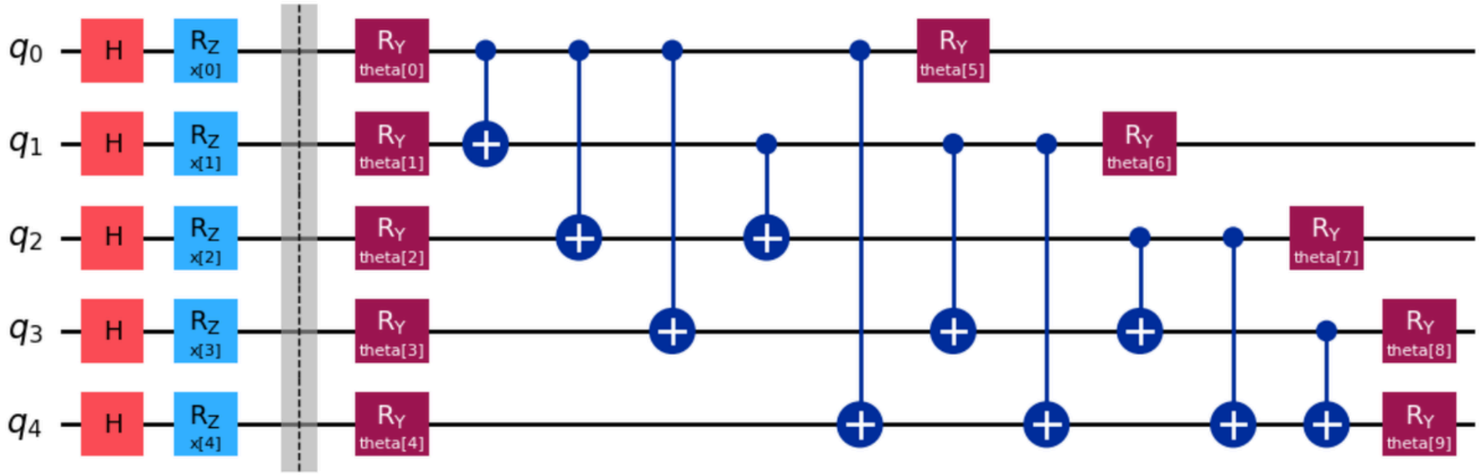
Theorem 2 (VC dimension lower bound for quantum circuits). *For d-dimensional input space, there is a quantum circuit hypothesis set with VC dimension* $d_{VC} \geqslant 2d + 1$ *using d qubits, d encoding gates, and d variational gates.*

For a perceptron the upper bound is d+1

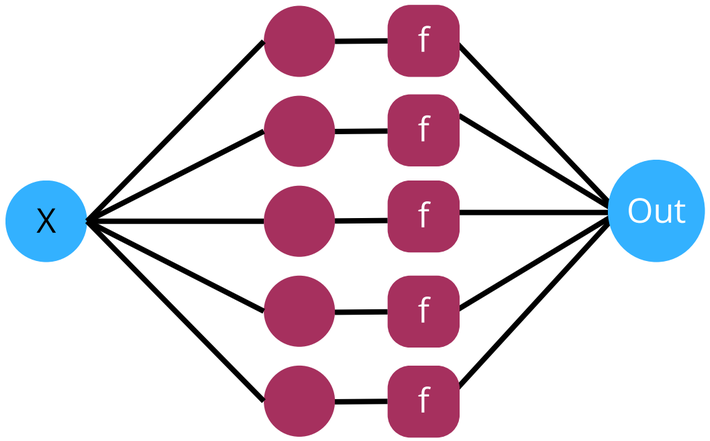
So is the quantum neuron more powerful?

- That VC dim is obtained by giving a parameter dependent linear feature map
- If we give a classical perceptron a sinusoidal feature map it can obtain the same result
- Then for classical feature maps the power is the same

How much the entanglement affects the QNN power?

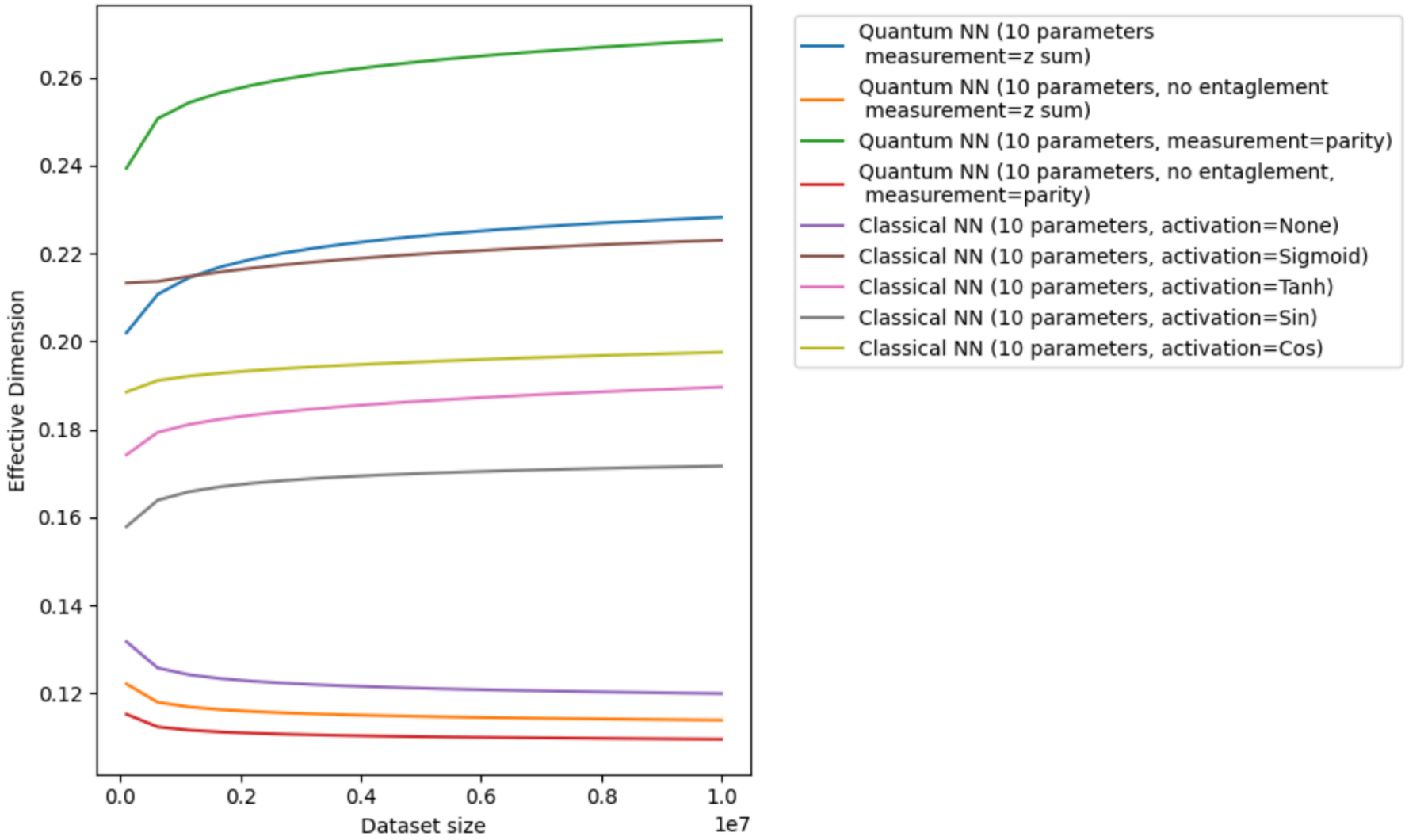


Where we rotate by the same x all the Qubits



Where f is the activation function

How much the entanglement affects the QNN power?



Can we conclude that the QNN creates more powerfull feature maps?

Computing the final state for two qubits

$$|00\rangle \xrightarrow{H\otimes H} \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$R_z(x) = \begin{pmatrix} e^{-ix/2} & 0 \\ 0 & e^{ix/2} \end{pmatrix}$$

$$\frac{1}{2}\left(e^{-ix}|00\rangle + |01\rangle + |10\rangle + e^{ix}|11\rangle\right)$$

Computing the final state for two qubits

$$R_y(\theta_1) \otimes R_y(\theta_2)$$

We obtain $|\psi_3\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$, where:

$$a_{00} = \frac{e^{-ix}}{2}\left[\left(e^{ix}\sin\left(\frac{\theta_1}{2}\right)\sin\left(\frac{\theta_2}{2}\right) - \sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)e^{ix} + \cos\left(\frac{\theta_1}{2}\right)\cos\left(\frac{\theta_2}{2}\right)\right]$$

$$a_{01} = \frac{e^{-ix}}{2}\left[\left(-e^{ix}\sin\left(\frac{\theta_1}{2}\right)\cos\left(\frac{\theta_2}{2}\right) + \cos\left(\frac{\theta_1+\theta_2}{2}\right)\right)e^{ix} + \sin\left(\frac{\theta_2}{2}\right)\cos\left(\frac{\theta_1}{2}\right)\right]$$

$$a_{10} = \frac{e^{-ix}}{2}\left[\left(-e^{ix}\sin\left(\frac{\theta_2}{2}\right)\cos\left(\frac{\theta_1}{2}\right) + \cos\left(\frac{\theta_1+\theta_2}{2}\right)\right)e^{ix} + \sin\left(\frac{\theta_1}{2}\right)\cos\left(\frac{\theta_2}{2}\right)\right]$$

$$a_{11} = \frac{e^{-ix}}{2}\left[\left(e^{ix}\cos\left(\frac{\theta_1}{2}\right)\cos\left(\frac{\theta_2}{2}\right) + \sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)e^{ix} + \sin\left(\frac{\theta_1}{2}\right)\sin\left(\frac{\theta_2}{2}\right)\right]$$

Computing the final state for two qubits

The CNOT swipes the amplitudes of $|10\rangle$ and $|11\rangle$:

$$\begin{aligned} b_{00} &= a_{00} \\ b_{01} &= a_{01} \\ b_{10} &= a_{11} \\ b_{11} &= a_{10} \end{aligned}$$

Then, after the CNOT:

$$|\psi_4\rangle = b_{00}|00\rangle + b_{01}|01\rangle + b_{10}|10\rangle + b_{11}|11\rangle$$

Computing the final state for two qubits

$$c_3 = \cos\left(\frac{\theta_3}{2}\right), \quad s_3 = \sin\left(\frac{\theta_3}{2}\right)$$
$$c_4 = \cos\left(\frac{\theta_4}{2}\right), \quad s_4 = \sin\left(\frac{\theta_4}{2}\right)$$

$$\begin{aligned}
\psi_{00} &= c_3 c_4 b_{00} - s_3 c_4 b_{10} - s_4 c_3 b_{01} + s_3 s_4 b_{11} \\
\psi_{01} &= s_4 c_3 b_{00} + c_3 c_4 b_{01} - s_3 s_4 b_{10} - s_3 c_4 b_{11} \\
\psi_{10} &= s_3 c_4 b_{00} - s_3 s_4 b_{01} + c_3 c_4 b_{10} - c_3 s_4 b_{11} \\
\psi_{11} &= s_3 s_4 b_{00} + s_3 c_4 b_{01} + c_3 s_4 b_{10} + c_3 c_4 b_{11}
\end{aligned}$$

$$|\psi_f\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$$

Computing the final state for two qubits

$$c_3 = \cos\left(\frac{\theta_3}{2}\right), \quad s_3 = \sin\left(\frac{\theta_3}{2}\right)$$
$$c_4 = \cos\left(\frac{\theta_4}{2}\right), \quad s_4 = \sin\left(\frac{\theta_4}{2}\right)$$

$$\begin{aligned}
\psi_{00} &= c_3 c_4 b_{00} - s_3 c_4 b_{10} - s_4 c_3 b_{01} + s_3 s_4 b_{11} \\
\psi_{01} &= s_4 c_3 b_{00} + c_3 c_4 b_{01} - s_3 s_4 b_{10} - s_3 c_4 b_{11} \\
\psi_{10} &= s_3 c_4 b_{00} - s_3 s_4 b_{01} + c_3 c_4 b_{10} - c_3 s_4 b_{11} \\
\psi_{11} &= s_3 s_4 b_{00} + s_3 c_4 b_{01} + c_3 s_4 b_{10} + c_3 c_4 b_{11}
\end{aligned}$$

$$|\psi_f\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$$

The last layer of rotation does not behave as a classical perceptron

Measurement

$$\langle O \rangle = \text{Sign}\left(|\psi_{00}|^2 + |\psi_{11}|^2 - |\psi_{01}|^2 - |\psi_{10}|^2 \right)$$

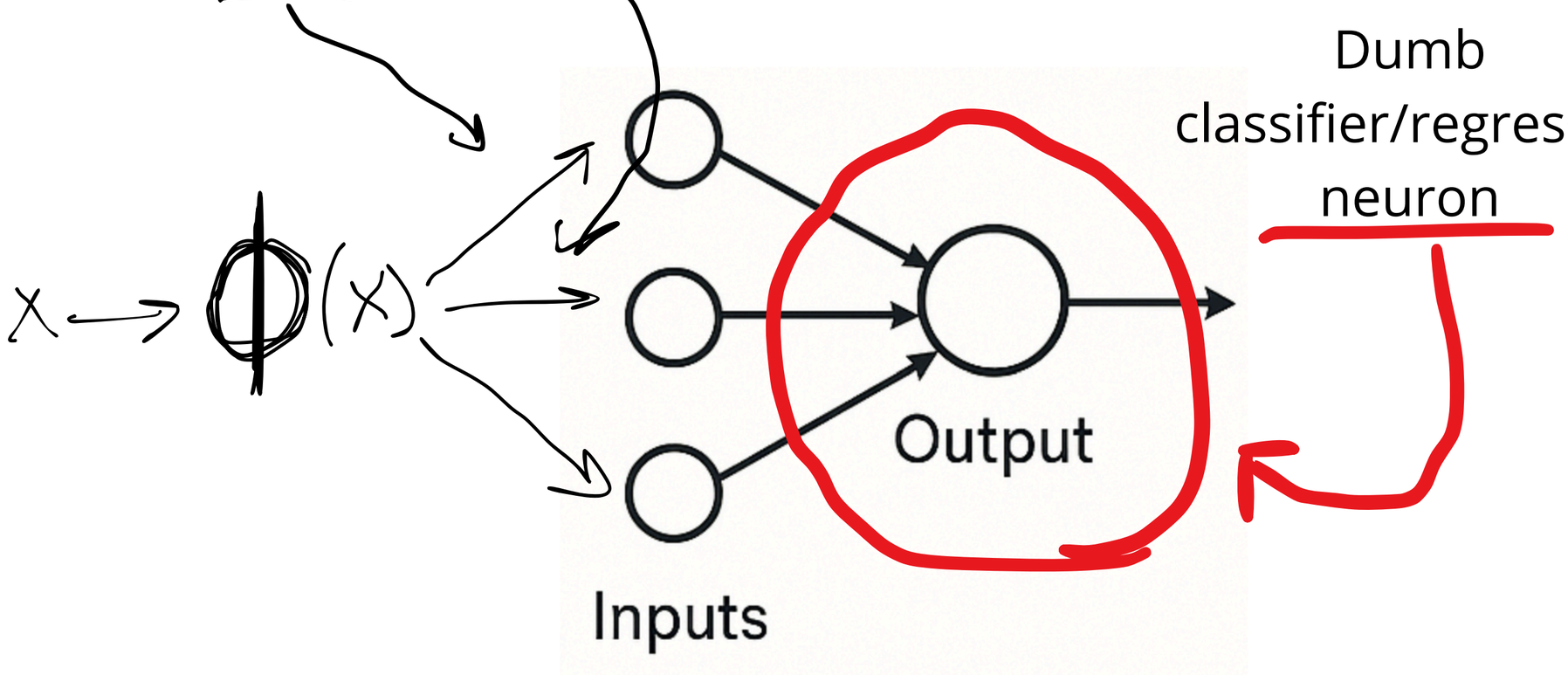
$$\langle O \rangle = \text{Sign}\left(w \cdot x \right)$$

Where $x = (|\psi_{00}|^2, \ldots, |\psi_{11}|^2)$ and $w = (1, -1, -1, 1)$

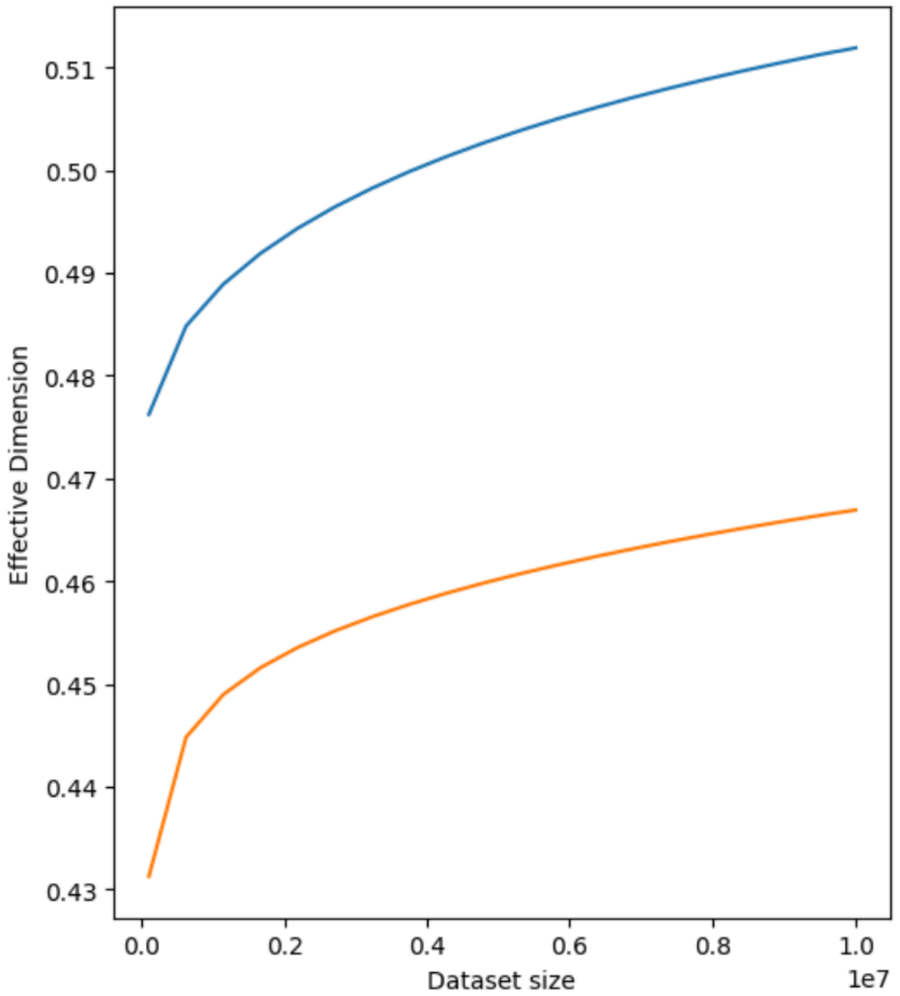
Definition of a new schema for a QNN

$$\langle O \rangle = \mathrm{Sign}\left(w \cdot x\right)$$

Where $x = (|\psi_{00}|^2, \ldots, |\psi_{11}|^2)$ and $w = (1, -1, -1, 1)$

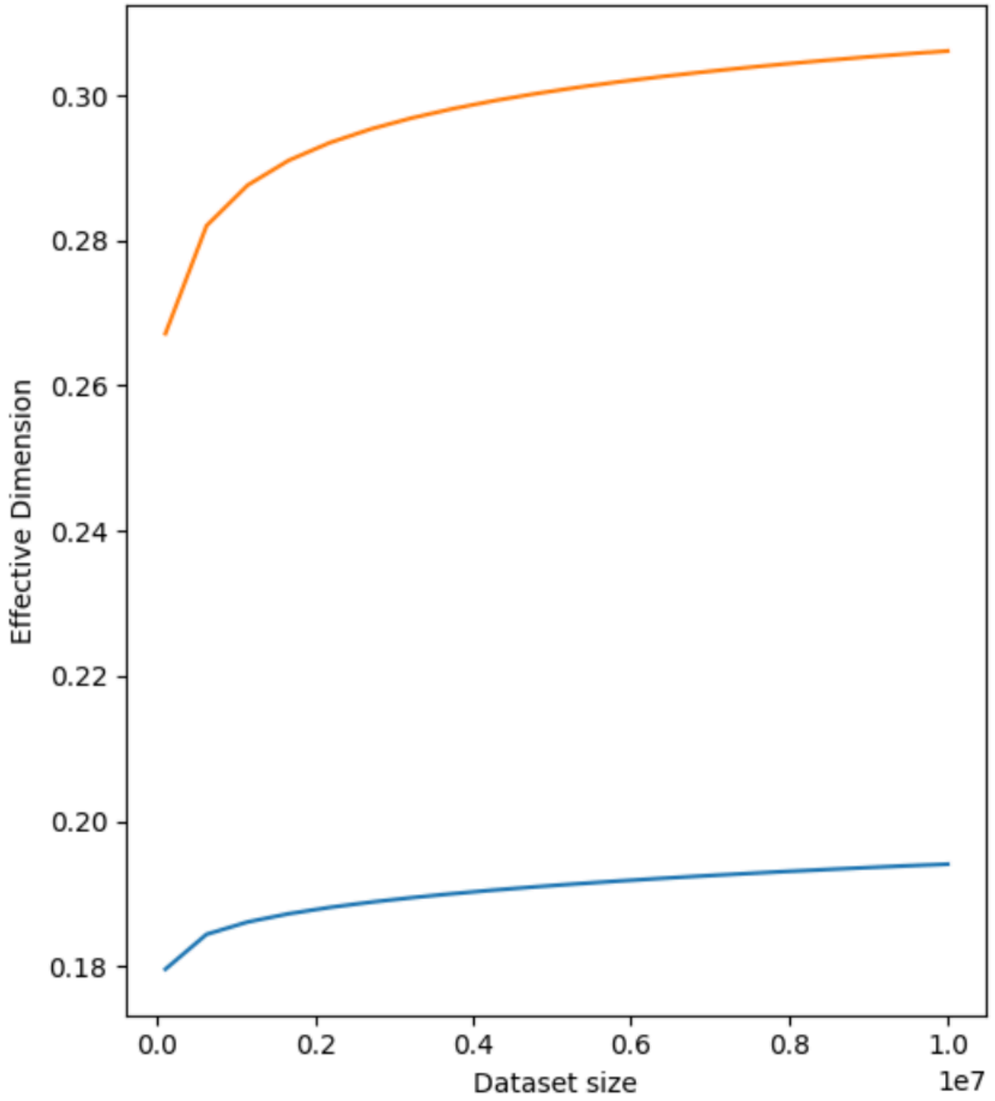


Result



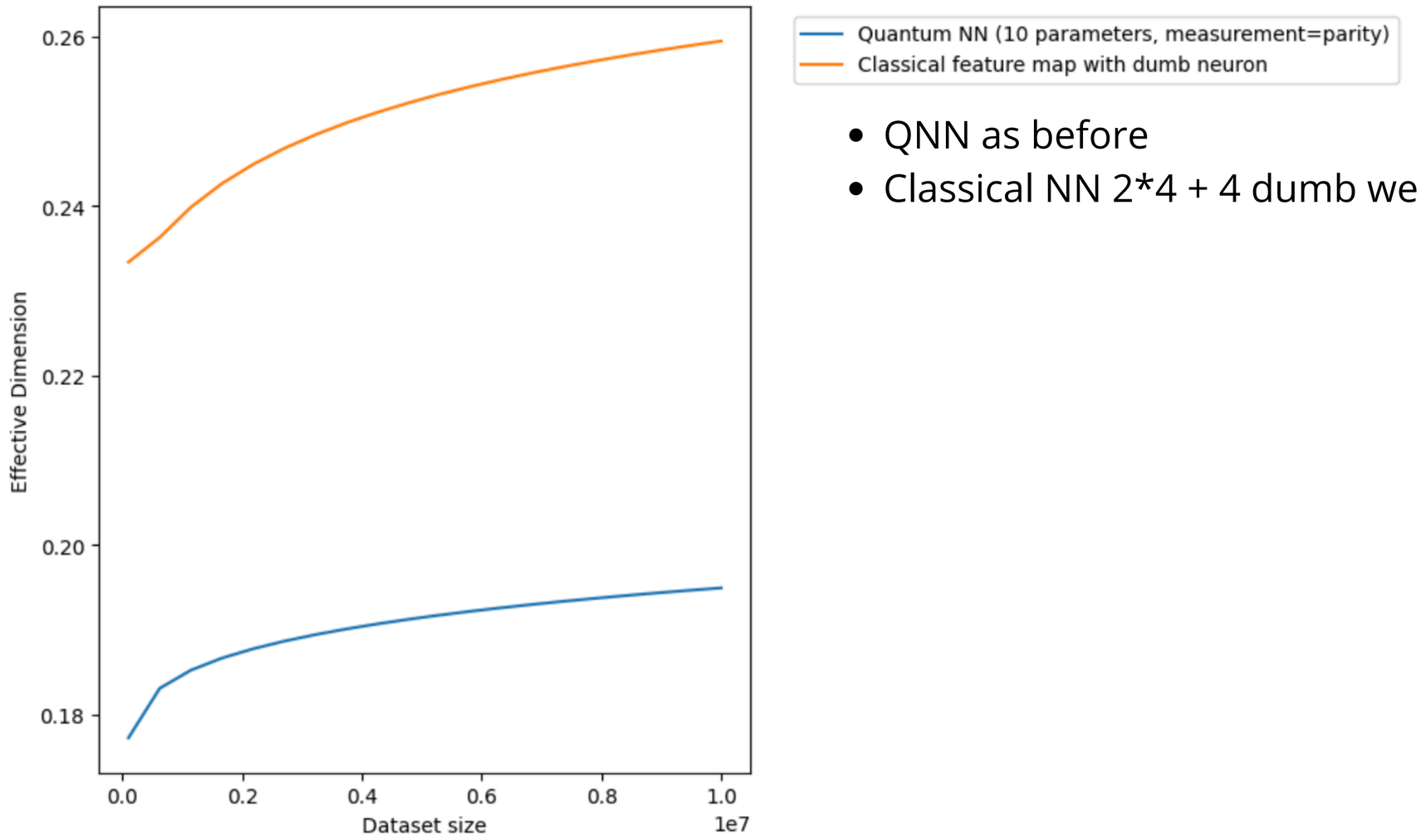
- QNN as before
- Classical NN 1 layer of 4 and 4 dumb weights

Result



- QNN as before
- Classical NN 2*4 + 4 dumb weights

Result



- QNN as before
- Classical NN 2*4 + 4 dumb weights

Appendix

Neuron’s effective dimension

Quantum Neuron

State setting:

$$\hat{x} = \frac{x}{x_{max} + x_{min}}$$

$$\tilde{x} = \cos^{-1}(\hat{x})$$

$$H|0\rangle = |+\rangle_x$$

$$R_z(\tilde{x})|+\rangle = |\psi\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\tilde{x}}|1\rangle\right)$$

The effect of the neuron:

Neuron’s effective dimension

The effect of the neuron:

$$R_y(\theta)|\psi\rangle = \frac{1}{\sqrt{2}}\left((\cos\frac{\theta}{2} - e^{i\tilde{x}}\sin\frac{\theta}{2})|0\rangle + (\cos\frac{\theta}{2} + e^{i\tilde{x}}\sin\frac{\theta}{2})|1\rangle\right)$$

Derivation for the probability of measuring the state $|0\rangle$:

$$p_0 = \frac{1}{2}(\cos\frac{\theta}{2} - e^{i\tilde{x}}\sin\frac{\theta}{2})(\cos\frac{\theta}{2} - e^{-i\tilde{x}}\sin\frac{\theta}{2})$$

$$p_0 = \frac{1}{2}(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} - e^{-i\tilde{x}}\cos\frac{\theta}{2}\sin\frac{\theta}{2} - e^{i\tilde{x}}\cos\frac{\theta}{2}\sin\frac{\theta}{2})$$

$$p_0 = \frac{1}{2}(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} - \cos\tilde{x}\sin\theta)$$

$$p_0 = \frac{1 - \cos\tilde{x}\sin\theta}{2}$$

Neuron’s effective dimension

The same derivation can be done for the probability of measuring the state $|1\rangle$:

$$p_1 = \frac{1 + \cos \tilde{x} \sin \theta}{2}$$

So the probability of measuring k times the state $|0\rangle$ in n measuments is:

$$p(k) = \frac{n!}{(n-k)!k!} p_0^k p_1^{n-k}$$

We can define the output function of the neuron as:

$$f(\hat{x}) = 2(\frac{k}{n} - \frac{1}{2})$$

Neuron’s effective dimension

So the distribution of $f(x)$ is obtained as:

$$k = \frac{n}{2}(f(x)+1)$$

$$p_f(f) = p_k(k(f))$$

$$p_f(f) = \frac{n!}{[n-\frac{n}{2}(f(x)+1)]![\frac{n}{2}(f(x)+1)]!} p_0^{\frac{n}{2}(f(x)+1)} p_1^{n-\frac{n}{2}(f(x)+1)}$$

$$p_f(f) = \frac{n!}{[\frac{n}{2}(1-f(x))]![\frac{n}{2}(f(x)+1)]!} p_0^{\frac{n}{2}(f(x)+1)} p_1^{\frac{n}{2}(1-f(x))}$$

Neuron’s effective dimension

Using the likelihood is:

$$p(y|x,\theta,\sigma_y) = p(y|f(x),\sigma_y)$$

But $f(x)$ is a random variable, we should evaluate $\langle f \rangle(x)$:

$$\langle f \rangle(\hat{x}) = 2(\frac{\langle k \rangle}{n} - \frac{1}{2}) = 2(\frac{np_0}{n} - \frac{1}{2}) = -\cos\tilde{x}\sin\theta = \hat{x}\sin\theta$$

$$Var(\langle f \rangle) = \frac{4}{n^2}Var(k) = \frac{4}{n^2}np_0p_1 = \frac{1-\cos^2\tilde{x}\sin^2\theta}{n} = \frac{1-\hat{x}^2\sin^2\theta}{n}$$

So we can define an effective error for the likelihood:

$$\sigma_{eff} = \sigma_y^2 + \sigma_f^2$$

Neuron’s effective dimension

The likelihood can be expressed as:

$$p(y|x,\theta,\sigma_y) = \mathcal{N}_y(\langle f\rangle(x), \sigma_y^2)p(f(x))$$

Then:

$$\log p(y|x,\theta,\sigma_y) = -\frac{1}{2}\log(2\pi\sigma_{eff}^2) - \frac{(y-(\langle f\rangle(x))^2}{2\sigma_{eff}^2}$$

$$\frac{\partial}{\partial\theta}\langle f\rangle(\theta) = \hat{x}\cos\theta$$

$$\frac{\partial}{\partial\theta}\sigma_{eff}^2(\theta) = -\frac{\hat{x}^2\sin(2\theta)}{n} = -\frac{2\hat{x}^2\sin\theta\cos\theta}{n}$$

$$I(\theta \mid \hat{x}) = \frac{\hat{x}^2\cos^2\theta}{\sigma_{eff}^2} + \frac{1}{2\sigma_{eff}^4}\left(\frac{2\hat{x}^2\sin\theta\cos\theta}{n}\right)^2$$

$$I(\theta \mid \hat{x}) = \frac{\hat{x}^2\cos^2\theta}{\sigma_{eff}^2(\hat{x})} + \frac{2\hat{x}^4\sin^2\theta\cos^2\theta}{n^2\,\sigma_{eff}^4(\hat{x})}$$

Neuron’s effective dimension

Classical Neuron

$$f(x;w) = wx$$

$$p(y \mid x;\theta) = \mathcal{N}(y;\, f(x;\theta),\, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - f(x;\theta))^2}{2\sigma^2}\right)$$

$$\log p(y \mid x;\theta) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(y-f)^2}{2\sigma^2}$$

$$\frac{\partial \log p}{\partial w} = \frac{(y-f)}{\sigma^2}\, x$$

$$(\frac{\partial \log p}{\partial w})^2 = \frac{(y-f)^2}{\sigma^4}\, x^2$$

$$\mathbb{E}_y[(y-f)^2] = \mathrm{Var}(y) = \sigma^2$$

$$I(\theta|x) = \frac{Var(x)}{\sigma^2}$$

Neuron’s effective dimension

Generalizing in multiple dimensions

Classical Neuron

$$\frac{\partial \log p}{\partial w_i} = \frac{(y-f)}{\sigma^2} x_i \quad \Rightarrow \quad I_{ij}(\mathbf{w}) = \mathbb{E}_y \left[\frac{\partial \log p}{\partial w_i} \frac{\partial \log p}{\partial w_j} \right] = \frac{x_i x_j}{\sigma^2}$$

$$I(\mathbf{w} \mid \mathbf{x}) = \frac{1}{\sigma^2} \mathbf{x}\mathbf{x}^\top$$

Quantum Neuron

$$\langle f \rangle(\mathbf{x}; \theta) = \sum_i \hat{x}_i \sin\theta_i$$

$$\frac{\partial \langle f \rangle}{\partial \theta_i} = \hat{x}_i \cos\theta_i$$

$$\frac{\partial \sigma^2_{eff}}{\partial \theta_i} = -\frac{2\hat{x}_i^2 \sin\theta_i \cos\theta_i}{n}$$

$$I_{ij}(\theta) = \frac{1}{\sigma^2_{eff}} \frac{\partial \langle f \rangle}{\partial \theta_i} \frac{\partial \langle f \rangle}{\partial \theta_j} + \frac{1}{2\sigma^4_{eff}} \frac{\partial \sigma^2_{eff}}{\partial \theta_i} \frac{\partial \sigma^2_{eff}}{\partial \theta_j}$$

Neuron’s effective dimension

Comparison

$$I^{classic} = \frac{1}{\sigma^2}\,\mathbf{x}\mathbf{x}^\top, \qquad I^{quantum} = \frac{1}{\sigma_{eff}^2}\,\mathbf{u}\mathbf{u}^\top + \frac{1}{2\sigma_{eff}^4}\,\mathbf{v}\mathbf{v}^\top$$

where

$$u_i = \hat{x}_i \cos\theta_i, \qquad v_i = \frac{2\hat{x}_i^2 \sin\theta_i \cos\theta_i}{n}$$

Neuron’s effective dimension

For $n >> 1$ we have:

$$I^{classic} = \frac{1}{\sigma^2}\mathbf{x}\mathbf{x}^\top, \qquad I^{quantum} = \frac{1}{\sigma_y^2}\mathbf{u}\mathbf{u}^\top$$

For Appendix B.3 Abbas et al.

$$d_{\gamma,n} = \bar{r} \qquad n \to \infty$$

Where $\bar{r} = \max_{\theta\in\Theta} Rank(I(\theta))$

Since the max rank for $I^{quantum}$ is d is reached for $\theta_i = 2n\pi$ for any $i \in 1, 2, \ldots, d$ we conclude that both the neurons have same effective dimension.

Metric in the state space

QNN parametrized by: $\theta_1, .., \theta_d$ The circuit produces: $|\psi(\theta) >= U(\theta)|\psi(0) > \in \mathcal{H}$

The measure is a projection over the Z axis, so two states that only differ by a phase have the same measure

The projective space $P(\mathcal{H})$ is a differentiable manifold $|\partial_i \psi >:= \partial_{\theta_i} |\psi(\theta) >$

We can define the tensor: $Q_{ij} =< d_i\psi|d_j\psi > - < d_i\psi|\psi >< \psi|d_j\psi >$

$G_{ij} = Re(Q_{ij})$ is a metric which measures the distance between nearby quantum states.

A larger value of G indicates that two states become rapidly distinguishable in the projective space.

Metric in the state space

We can use G to get advantage from using a qnn:

$$\theta^{n+1} = \theta^{n} - \eta G^{-1} \nabla L$$

Taking in account the geometrci proprieties of the quantumstate space